

Development of the thorax

THORACIC WALL AND DIAPHRAGM

The thorax as an entity is not apparent in embryos until the end of the embryonic stage of development (stage 23). It develops around the early pericardial cavity and the associated pericardioperitoneal canals (Fig. 52.1A; see Fig. 52.3A). The pericardioperitoneal canals give rise to the pleural cavities surrounding the lungs and that part of the peritoneal cavity surrounding the lower end of the foregut, which becomes the distal oesophagus, stomach and proximal duodenum. Later, these portions of peritoneal cavity are sequestered into the abdominal cavity by the development of the diaphragm.

In stage 14 embryos, the heart is at the level of the upper cervical somites and above the upper limb buds. The thoracic somites are opposite the midgut. The putative thoracic region contains the pericardial cavity ventrally and the pericardioperitoneal canals posteriorly, on each side of the foregut. The future pleural cavities are as yet undefined regions of the pericardioperitoneal canals. Below the heart, septum transversum mesenchyme is being invaginated by endodermal epithelial cells from the foregut hepatic primordium.

The heart and pericardial cavity are relatively large in the early embryo (Fig. 52.1B; see Fig. 52.3A). Throughout development, the lungs remain unexpanded and do not achieve their full size within the thorax, reflecting the fact that the placenta, and not the lungs, is the organ of fetal respiration. The lower partition of the thorax, the diaphragm, can be identified in stage 13 embryos; it migrates caudally

in line with the craniocaudal progression of development and the elongation of the neck. The parietal pericardium remains attached to the diaphragm as it descends.

For the lungs to inflate and deflate, they must be surrounded by a complete pleural cavity slightly larger than the capacity of the lungs. The synchronous development of the thoracic cage, diaphragm and pleural cavities is, therefore, of vital importance for the normal development of the lungs and postnatal functioning of the respiratory system. Before birth, fetal breathing movements may be observed using ultrasonography. These can have no respiratory function, but it is thought that these phasic changes lead to the release of growth factors that promote normal lung development. These movements require an intact thoracic and abdominal cavity, so that the diaphragm can descend towards the rigid pelvic bowl, pushing out the abdominal wall. In conditions where this is not the case, e.g. exomphalos, the lungs may be smaller than normal. Other causes of interference with these movements include neuromuscular disorders of antenatal onset, such as severe spinal muscular atrophy and myotonic dystrophy inherited from the mother, and reduced volume of amniotic fluid, e.g. in Potter's syndrome (bilateral renal agenesis, oligohydramnios sequence).

The lung buds are invested by splanchnopleuric mesenchyme derived from the medial walls of the pericardioperitoneal canals, whereas the lateral walls produce somatopleuric mesenchyme, which contributes to the body wall. This latter mesenchyme is penetrated by the developing ribs, which arise from the thoracic sclerotomes. In the midline, the somatopleuric mesenchyme gives rise to the sternum and

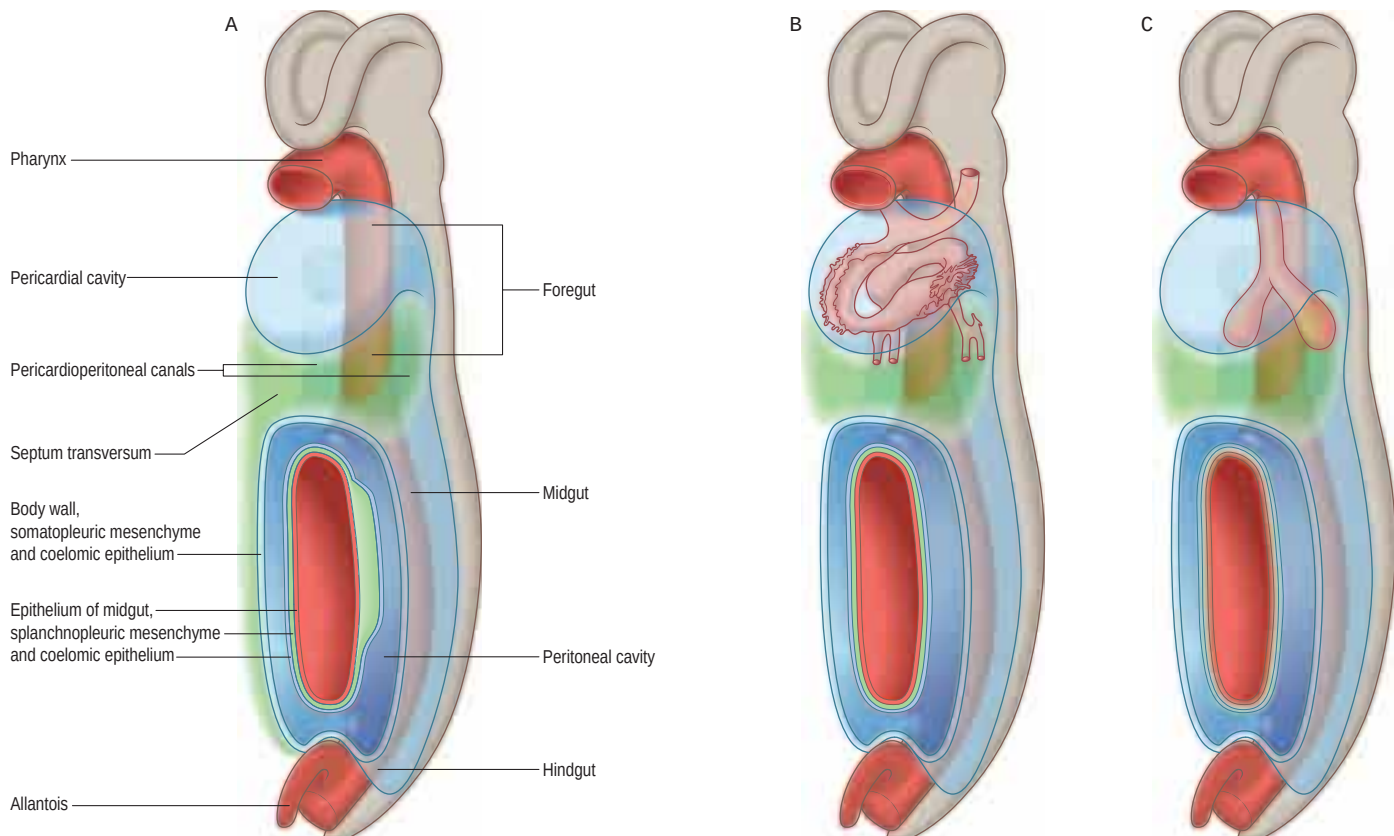


Fig. 52.1 **A**, The arrangement of the epithelial populations in an embryo at about stage 11. The future thoracic region lies caudal to the pharynx and extends to the upper part of the septum transversum. **B**, The heart is shown *in situ* within the pericardial cavity. Venous tributaries return blood from the body, yolk sac and placenta, and arterial trunks pass each side of the pharynx. **C**, The respiratory diverticulum arises in the midline from the lower portion of the pharynx, and lung buds extend dorsally into the pericardioperitoneal canals.

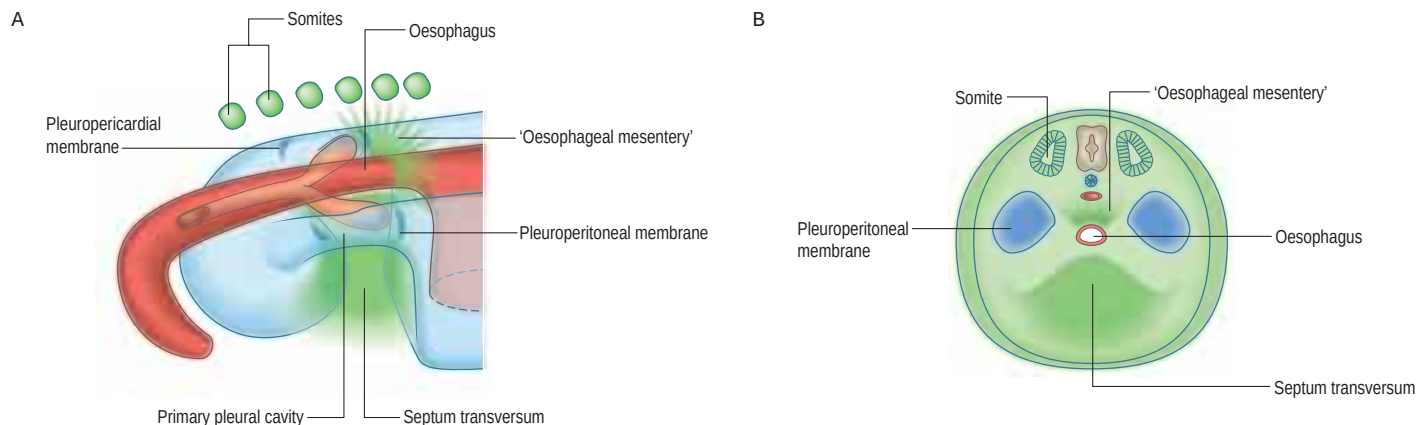


Fig. 52.2 **A**, A dorsolateral view of the foregut between the pericardioperitoneal canals. The respiratory diverticulum arises from the ventral side of the pharynx and grows caudally. Bilateral lung buds extend dorsally into the pericardioperitoneal canals dorsolaterally. Later changes in position of the common cardinal veins contribute to the formation of the pericardiopleural membranes cranial to the developing lungs, whereas folds of coelomic epithelium, pleuroperitoneal membranes, separate the pleural cavities from the peritoneal cavity. **B**, Components of the diaphragm.

costal cartilages. The bony and cartilaginous cage provides insertions for the intercostal muscles, which arise from the ventrolateral edge of the epithelial plate of the somites. The somatopleuric coelomic epithelium, after its proliferative phase, contributes to the mesothelium of the parietal layer of pleura.

As the lung buds project into the pericardioperitoneal canals (Figs 52.1C, 52.2A), they subdivide them into primary pleural coeloms around the lung buds cranially, and paired peritoneal coeloms caudally, which are contiguous with the wider peritoneal coelom around the mid- and hindguts. The communications with the pericardial and peritoneal coeloms become termed the pleuropericardial and pleuroperitoneal canals, respectively. When separation between these fluid-filled major coelomic regions is advancing towards completion, they are named the pericardial, pleural and peritoneal cavities. In early embryos, the cavities retain substantial volumes of fluid and their walls are separate; they provide the route for a primitive type of circulation until superseded by the blood vascular system. In later fetal and postnatal life, the cavity walls are coapted, so that a mere microscopic film of serous fluid intervenes between them.

A curved elevation of tissue, the pulmonary ridge, develops on the lateral wall of the pleural coelom and partly encircles the pleuropericardial canal. The ridge is continuous with the dorsolateral edge of the septum transversum. The developing lung bud abuts on the ridge, which, as a result, divides into two diverging membranes meeting at the septum transversum. One is cranially placed and termed the pleuropericardial membrane (see Fig. 52.2A); embedded within it are the common cardinal vein and phrenic nerve, which reach the septum transversum by this route. The other membrane, caudally placed, is termed the pleuroperitoneal membrane (see Fig. 52.2A). As the apical part of the lung forms, it invades and splits the body wall, and extends cranially on the lateral aspect of the common cardinal vein, preceded by an extension from the primary pleural coelom to form part of the secondary, definitive, pleural sac. In this way, the common cardinal vein and the phrenic nerve come to lie medially in the mediastinum. The pleuropericardial canal, which lies medial to the vein, is gradually narrowed to a slit, which is soon obliterated by the apposition and fusion of its margins. Closure occurs early and is mainly effected by the growth and expansion of the surrounding viscera (heart and great vessels, lungs, trachea and oesophagus), and not by active growth of the pleuropericardial membrane across the opening to the root of the lung.

In addition to its extension in a cranial direction, the lung and its associated visceral and parietal pleurae also enlarge ventromedially and caudodorsally. With the ventromedial extension, the lungs and pleurae therefore excavate and split the somatopleuric mesenchyme over the pericardium, separating the latter from the ventral and lateral thoracic walls (Fig. 52.3B–D). The ventrolateral fibrous pericardium, parietal serous pericardium and mediastinal parietal pleura, although topographically deep, are, therefore, somatopleuric in origin.

Congenital anomalies of the chest wall

The dorsal portion of the thorax is derived from somites, the sclerotomal portions of which form the thoracic vertebrae and ribs (Ch. 44). Vertebral anomalies, including formation of hemivertebrae and block vertebrae (where cranial and caudal sclerotomal halves do not separate), may contribute to scoliosis. Five per cent of scoliosis is congenital; it may be associated with multiple other anomalies, e.g. the VACTERL

(vertebral; anorectal; cardiac; tracheal; esophageal; renal; limb) association, or with congenital syndromes such as Marfan's and congenital neurofibromatosis. The diagnosis may be missed at birth. There are a number of syndromes in which the chest wall does not develop properly, and the lungs are consequently hypoplastic. These include Jeune's syndrome (asphyxiating thoracic dystrophy – an autosomal recessive condition that may be associated with short-limb dwarfism and polydactyly). Ribcage abnormalities also occur in thanatophoric dwarfism, achondroplasia, chondroectodermal dysplasia and giant exomphalos. Many of these conditions are fatal soon after birth; milder forms may improve with time, but affected individuals may need prolonged respiratory support.

DIAPHRAGM

The separation of the pleural and peritoneal cavities is effected by development of the diaphragm, which forms from a portion of the septum transversum mesenchyme above the developing liver (see Fig. 52.2). The septum transversum is a population of mesenchymal cells that arises from the coelomic wall of the caudal part of the pericardial cavity. As the population proliferates, it forms a condensation of mesenchyme, caudal to the pericardial cavity and extending from the ventral and lateral regions of the body wall to the foregut. Dorsal to it, on each side, are the relatively narrow pleuroperitoneal canals. The endodermal hepatic bud grows into the caudal part of the septum transversum, whereas the cranial portion will form the diaphragm.

The oesophagus and stomach are medial to the pleuroperitoneal canals. As development proceeds, the lower portion of the oesophagus inclines ventrally anterior to the descending thoracic aorta. Although the oesophagus has no true ventral or dorsal mesentery, in descriptions of diaphragmatic development, the portion of mesenchyme between the oesophagus and aorta at the level of the forming diaphragm is often homologized with part of a dorsal meso-oesophagus. The pleuroperitoneal membranes, which remain small, are dorsolateral to the canals, and the mesonephric ridges, suprarenal glands and gonads are dorsal. Just as the enlargement of the pleural cavity cranially and ventrally is effected by a process of burrowing into the body wall, so is its caudodorsal enlargement. The expanding pleural cavities extend into the mesenchyme dorsal to the suprarenal glands, the gonads and (degenerating) mesonephric ridges. Thus, somatopleuric mesenchyme is peeled off the dorsal body wall to form a substantial portion of the dorsolumbar part of the diaphragm (see Fig. 52.3C). The pleuroperitoneal canal is closed by the fusion of its edges, which are carried together from posterolaterally to anteromedially by growth of the organs surrounding it: in particular, by growth of the suprarenal gland. The right pleuroperitoneal canal closes earlier than the left, which presumably explains why an abnormal communication persisting between the pleural and peritoneal cavities is more frequently encountered on the left.

While these changes are occurring, the septum transversum undergoes a progressive alteration in relative position. The dorsal border of the septum transversum, which initially lies opposite the second cervical segment, migrates caudally as the embryo grows and the heart enlarges. At first, the ventral border moves more rapidly than the dorsal, but after the embryo has attained a length of 5 mm, the dorsal border

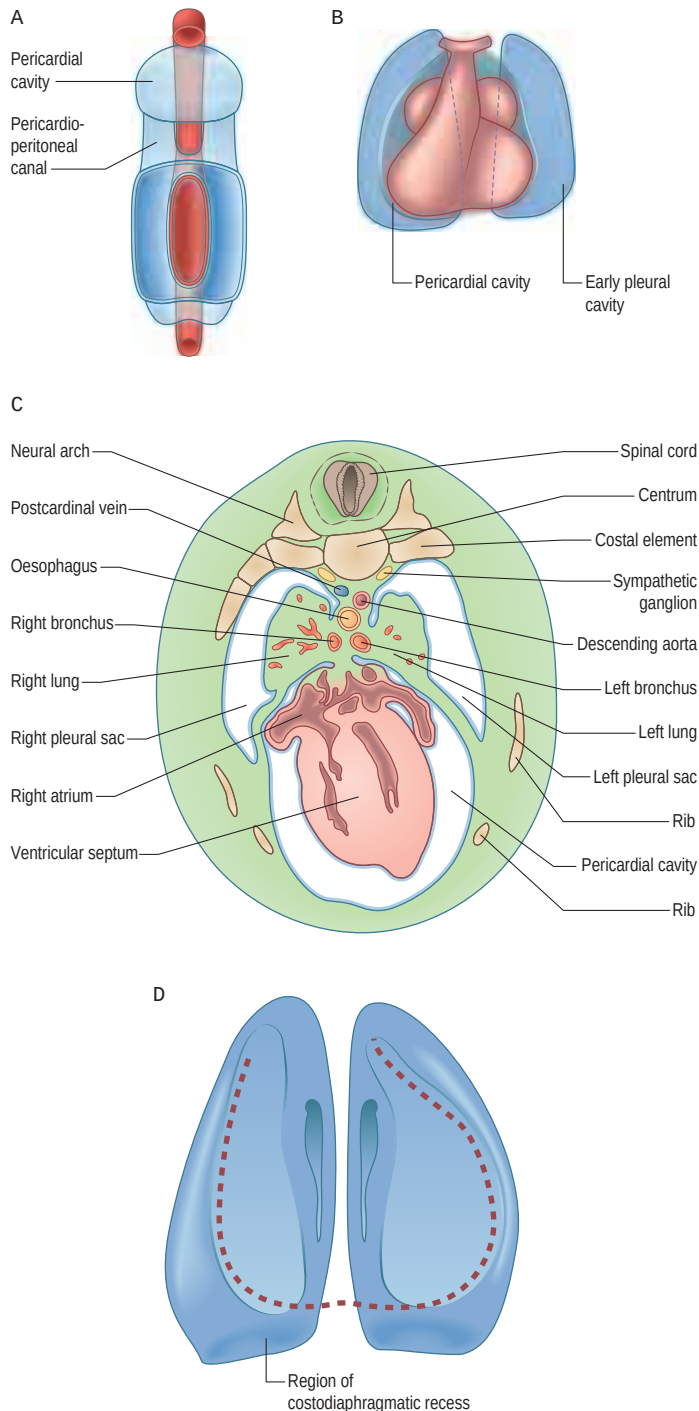


Fig. 52.3 **A**, A ventral view of the intraembryonic coelom forming early body cavities. **B**, The parietal pericardium is shown surrounding the developing heart ventrally, with the parietal pleura of the lungs dorsal and lateral on each side. **C**, A transverse section through the thoracic region, showing the ventral extensions of the pleural sacs on each side of the pericardium that split the body wall mesenchyme. **D**, An anterior view of the pleural sacs. The position of the pericardium is shown by a dotted line. Dorsolateral growth of the pleural sacs also splits the body wall dorsocaudally, defining the costodiaphragmatic recess.

migrates more rapidly. When the dorsal border of the septum transversum lies opposite the fourth cervical segment, the phrenic nerve (C3, 4 and 5) and portions of the corresponding myotomes grow into it and accompany it in its later migrations. The dorsal border of the septum transversum does not come to lie opposite the last thoracic and first lumbar segments, the final position occupied by some of the dorsal attachments of the diaphragm, until the end of the second month. The main derivatives of the central part of the diaphragm lie at considerably more cranial levels.

The diaphragm is, therefore, composed of the dorsal meso-oesophagus and paired pleuroperitoneal membranes (posteriorly); septum transversum mesenchyme (ventrally); and excavated body wall (posteriorly and laterally).

Diaphragmatic hernias

Diaphragmatic hernias may result from failure of fusion of the component parts or from a primary defect. Posterolateral defects (Bochdalek's hernia) are the most common (85–90%) and may be bilateral (5%) or unilateral. Of the unilateral defects, the left side is more commonly affected (80%). Although these hernias have been attributed to failure of fusion of the pleuroperitoneal membrane, there is increasing evidence that the primary abnormality is lung hypoplasia, and the herniation of the abdominal contents is secondary, which has important implications for treatment (Jesudason 2002). A non-muscular membranous sac, possibly derived from the pleuroperitoneal canal wall, is present in 10–15% of cases, signifying the early occurrence of this lesion prior to closure of the pleuroperitoneal canal. Hernias between the costal and sternal origins (Morgagni hernias) are rare (1–2%) (Fig. 52.4). Midline defects in the central tendon arise from septum transversum defects. The incidence of congenital diaphragmatic hernias is about 1:3000 to 1:5000 in neonates, with a prenatal incidence of 1:2000.

Diaphragmatic hernias can usually be diagnosed by antenatal ultrasound examination. The presence of bowel in the thorax and mediastinal shift may be seen sonographically and radiographically (Fig. 52.5). The main causes of death in babies with diaphragmatic hernias are pulmonary hypoplasia and hypertension. Pulmonary hypoplasia is related to the abnormal branching of the airways both ipsilaterally and contralaterally, resulting in the formation of reduced numbers of alveoli. Pulmonary hypertension is related to pulmonary hypoplasia, and also to the increased muscularization and reactivity of the pulmonary circulation. There may be associated abnormalities, including trisomy 13 and 18, and diaphragmatic hernia may be part of a number of syndromes, many of which may lead to termination of the pregnancy or to death of the neonate. For a review summarizing the recurrent genetic causes of congenital diaphragmatic hernia, see Wynn et al (2014).

Diaphragmatic eventration is a condition in which an abnormal diaphragmatic silhouette is seen on X-ray with an abnormal medial bulge into the thorax; it is associated with paradoxical motion of the diaphragm during respiratory movements. This may be a benign condition, often a chance finding and requiring no treatment, or it may lead to respiratory symptoms, requiring plication; it is occasionally associated with rib and cardiac abnormalities.

ESOPHAGUS

The development of the oesophagus is described in Chapter 36. Oesophageal atresia, with or without tracheo-oesophageal fistula, may present antenatally with polyhydramnios due to failure of movement of amniotic fluid into the gut after fetal swallowing (Ch. 9), and choking and inability to swallow saliva in the neonatal period.

Ultrasound antenatal imaging of the thorax

The overall shape of the fetal chest is best seen on coronal section. A bell-shaped chest is present in pulmonary hypoplasia. In axial section, the thorax is almost circular. Fetal chest size is estimated by measuring the chest circumference on axial section at the level of the four-chamber-heart view. Such measurement can confirm or exclude reduced chest volume (Karl and Chaoui 2015). The relative circumference of the thorax and heart remains the same in the second and third trimesters. The diaphragm is best viewed in sagittal view. Diaphragmatic hernias can be diagnosed from about 18 weeks onwards.

A number of differences between the neonatal and adult thorax and diaphragm are significant in early postnatal life, especially in the preterm infant. The full-term neonatal thorax has a rounded circumference, whereas the adult profile is dorsoventrally flattened. The compliance of the adult chest wall and lung is approximately equal, but the neonatal chest wall is up to five times more compliant than the lungs and so is easily deformable; indrawing of the chest wall is an important sign of increased work of breathing in the neonate. The diaphragm is relatively flat at birth and gains its dome shape as the thorax and abdominal viscera grow. The arrangement of the ribs and diaphragm in the neonate means that contraction of the diaphragm pulls the compliant ribs inwards; this does not contribute to inspiration and leads to chest distortion, and may be opposed by the stabilizing action of the expiratory muscles. The neonatal diaphragm exhibits an exaggerated asymmetric movement: the posterior portion shows a considerably greater excursion than the anterior portion.

A neonate undertakes relatively greater work in order to breathe, and is much more vulnerable to developing respiratory muscle fatigue and

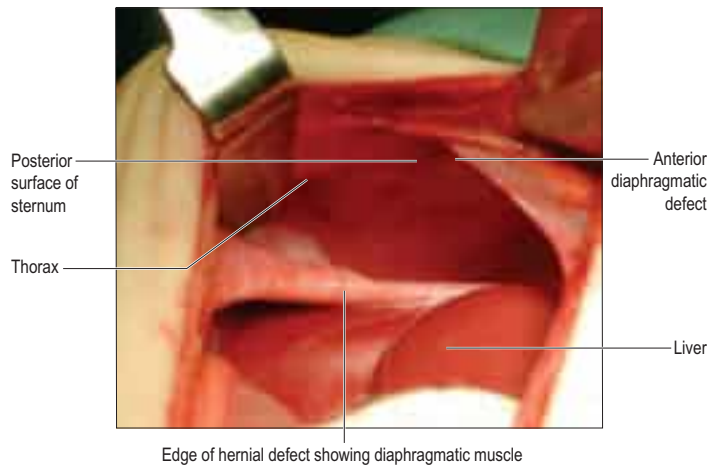


Fig. 52.4 A Morgagni hernia in an 8-year-old boy, as displayed at laparotomy. (Courtesy of Mr G Jawaheer, Great North Children's Hospital, Newcastle upon Tyne.)

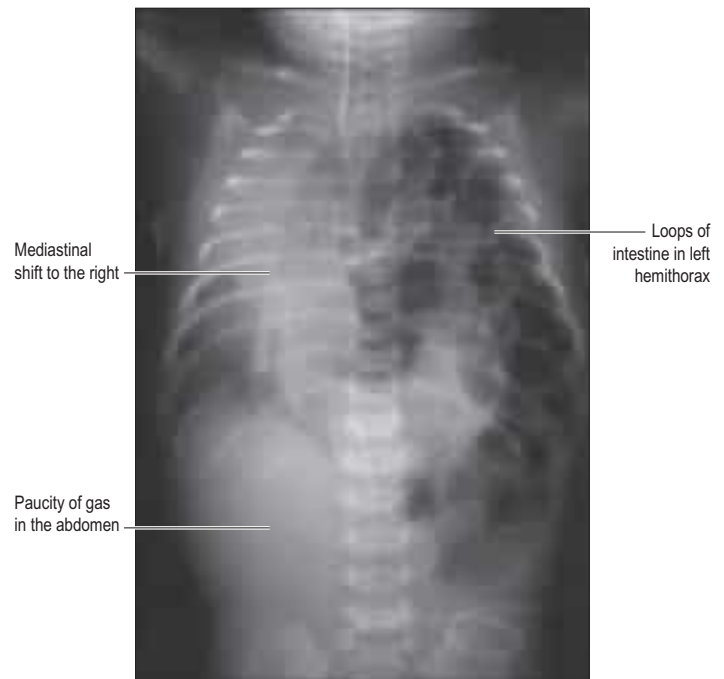


Fig. 52.5 A plain radiograph of the chest and abdomen in a 1-day-old neonate with left posterolateral congenital diaphragmatic hernia. (Courtesy of Mr G Jawaheer, Great North Children's Hospital, Newcastle upon Tyne.)

respiratory muscle failure than an adult. Experimental animal studies suggest that neonates may be especially vulnerable to respiratory muscle failure because of the initial immaturity of synapses. Moreover, myosin isoforms are developmentally regulated in the diaphragm; the reduced power output and work performance of the newborn diaphragm may reflect these molecular differences.

In expiration, there may be active braking of airflow, caused by inspiratory muscle activity and partial constriction of the larynx; this produces grunting on expiration, especially if the lungs are stiff, e.g. in neonatal respiratory distress. A very preterm neonate is difficult to study, but respiration is likely to be even more compromised by the compliant chest wall and lack of reserve than it is in the term newborn (van Vonderen et al 2014).

At all ages, there is a reduction in, if not loss of, tonic intercostal activity during rapid eye movement (REM) sleep. The mechanism is believed to be related to a descending spinal inhibition of the muscle spindle system. In addition, although, during REM sleep, the diaphragm descends further, this inspiratory effort is dissipated in sucking in the ribs and enlarging the abdomen; thus, the ribcage and abdominal ventilatory movements become out of phase. The neonate is at particular risk in this respect because the chest wall is flexible and much of the infant sleep activity is of the REM type. Furthermore, the upper airway musculature may lose tone during REM sleep, so that the soft tissues of the pharynx are sucked in during inspiration, limiting the cross-sectional area of the upper airway, and further increasing the work of breathing.

HEART AND GREAT VESSELS

In the last few decades, understanding of the development of the heart and great vessels has been revolutionized by molecular lineage tracing and cell division studies. Cardiac development has traditionally been taught from the standpoint of a formed heart tube, with venous and arterial ends, which folds within an external pericardial cavity. It is now clear that this view is incorrect. The cells contributing to the heart arise sequentially from a range of extracardiac cell populations, and local instructive interactions maintain the developmental progression; cardiomyocyte identity is not fixed but dependent on spatial and temporal signals (Sylva et al 2013). As the changing orientation of the developing heart within three dimensions is particularly difficult to understand conceptually, and considering that the received view of organogenesis is coloured by past explanations and specific pedagogic approaches, it is necessary to maintain some of the older terminology by which heart development was described, as well as considering the most recent findings.

Cardiac differentiation and morphogenesis are intimately related processes, achieved by a highly complicated transcriptional network. The cardiac transcription factor Nkx2-5, which has a homogeneous distribution over the embryonic heart tube, is generally important for cardiac-specific gene expression. In the human, mutations in this factor cause atrial septal defects and disturbances of atrioventricular conduction; in *Drosophila*, the heart will not form in its absence. The recent discovery of the involvement of the role of T-box transcription factors, which are locally expressed, provides some evidence of the further complexity of patterning the heart. Whereas Tbx5 confers caudocranial positional information over the heart tube, thereby determining the regional development of the cardiac chambers along these axes, Tbx2 and 3 prevent regional formation of the cardiac chambers, permitting the myocardium in these regions to differentiate into the conduction system. Tbx18 is involved in the development of the sinus venosus and Tbx1 in the development of the outflow tract of the heart.

A number of terms must be specified before development of the heart and discussion of the cardiac chambers can be described. The inflow tract of the heart starts at the transition of the systemic tributaries (sinus venosus) into the primary heart tube. Within the straight heart tube, forming atrial and ventricular chambers can be discerned, separated by an atrioventricular canal. The inflow tract thus connects the sinus venosus with the forming atrium, whereas the outflow tract connects the ventricles with the aortic sac; a proximal portion (close to the ventricle) and a distal portion can be defined within the outflow tract.

CELLS THAT GIVE RISE TO THE HEART

The heart is formed from tissues derived from the midline splanchnopleuric coelomic epithelium with later contributions from neural crest mesenchyme. The splanchnopleuric coelomic epithelium gives rise to the myocardium, including the conduction system of the heart, and the

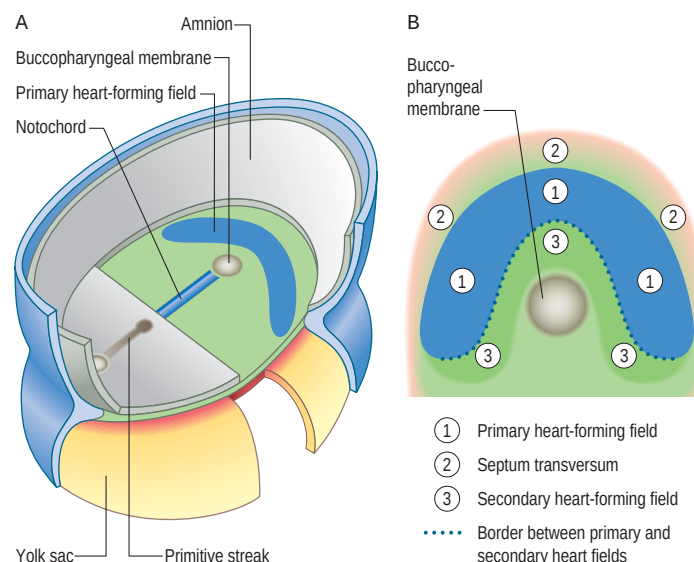


Fig. 52.6 **A**, The early embryonic disc viewed from a dorsolateral aspect, showing the position of the primary heart-forming field relative to the surface ectoderm of the trilaminar disc. **B**, The rostral end of the unfolded embryonic disc from a dorsal aspect, showing the position of the mesenchymal populations that will give rise to the primary heart-forming field, septum transversum mesenchyme and second heart-forming field.

endocardium, including its derived cardiac mesenchymal population, which produces the valvular tissues of the heart. Splanchnopleuric coelomic epithelium is also the source of the epicardium, coronary arteries and interstitial fibroblasts.

Primitive cardiac myocytes can first be seen in embryos at stage 8. During the onset of neurulation and somitogenesis, the intraembryonic coelom forms across the midline, initially above the endoderm, in a horseshoe-shaped area termed the primary heart-forming field (cardiac crescent) (Fig. 52.6). As the head fold emerges, the coelom undergoes a reversal, so that the future pericardial cavity comes to lie ventral to the endodermal foregut (see Fig. 52.1; Fig. 52.7). The splanchnopleuric wall of the pericardial coelom, subjacent to the endoderm, provides a germinal epithelium that produces early cardiac myocytes. It is characterized by the expression of myocardial-specific markers, such as cardiac myosin heavy chain and the cardiac transcription factor Nkx2-5. Later myocardial additions from mesenchyme localized central and peripheral to the cardiac crescent on the embryonic disc are termed the second heart-forming field (see Fig. 52.6). Whether these two myocardial fields represent distinct myocardial lineages, or constitute a single field, which is gradually added to the heart, remains a matter of debate.

The endocardium also develops during stage 8–9 from the coelomic splanchnopleuric epithelium. Cells arise singly close to the ventrolateral edges of the cranial intestinal portal and form an endocardial plexus between the splanchnopleuric coelomic epithelium and the foregut endoderm. These groups of cells are now termed angioblastic mesenchyme; they are amongst the earliest intraembryonic vascular precursors to appear and express markers for the endothelial cell lineage. The cells aggregate to form an epithelium, the endocardium, which encloses small cavities that coalesce in the vicinity of the developing foregut to establish bilateral, hollow, endocardial tubes. The latter fuse across the midline progressively, commencing at the outflow tract, or arterial pole, and extending to the inflow tract, or venous pole (see Fig. 52.7). By stage 10, a single endocardial tube is present and is almost completely surrounded by myocardial cells. This arrangement of an outer myocardial sleeve containing an inner endocardial tube constitutes the primary heart tube. The inner endocardial epithelium induces the myocardial cells to synthesize specific extracellular matrix proteins, which form a fine extracellular reticulum that holds the endocardial tube apart from the developing myocardium. Close to the foregut endoderm, the myocardial cells at the reflections of the pericardial splanchnopleuric epithelium form the dorsal mesocardium, which may stabilize the developing endothelium and promote the fusion of the bilateral endocardial tubes. The dorsal mesocardium encompasses a mesenchymal population specifically referred to as mediastinal mesenchyme, and is contiguous with the splanchnopleuric mesenchyme surrounding the embryonic foregut.

Much later in heart development, neural crest cells, arising from the region between the otic vesicle and the caudal limit of somite three, grow into the outflow tract of the heart. They are believed to play a role

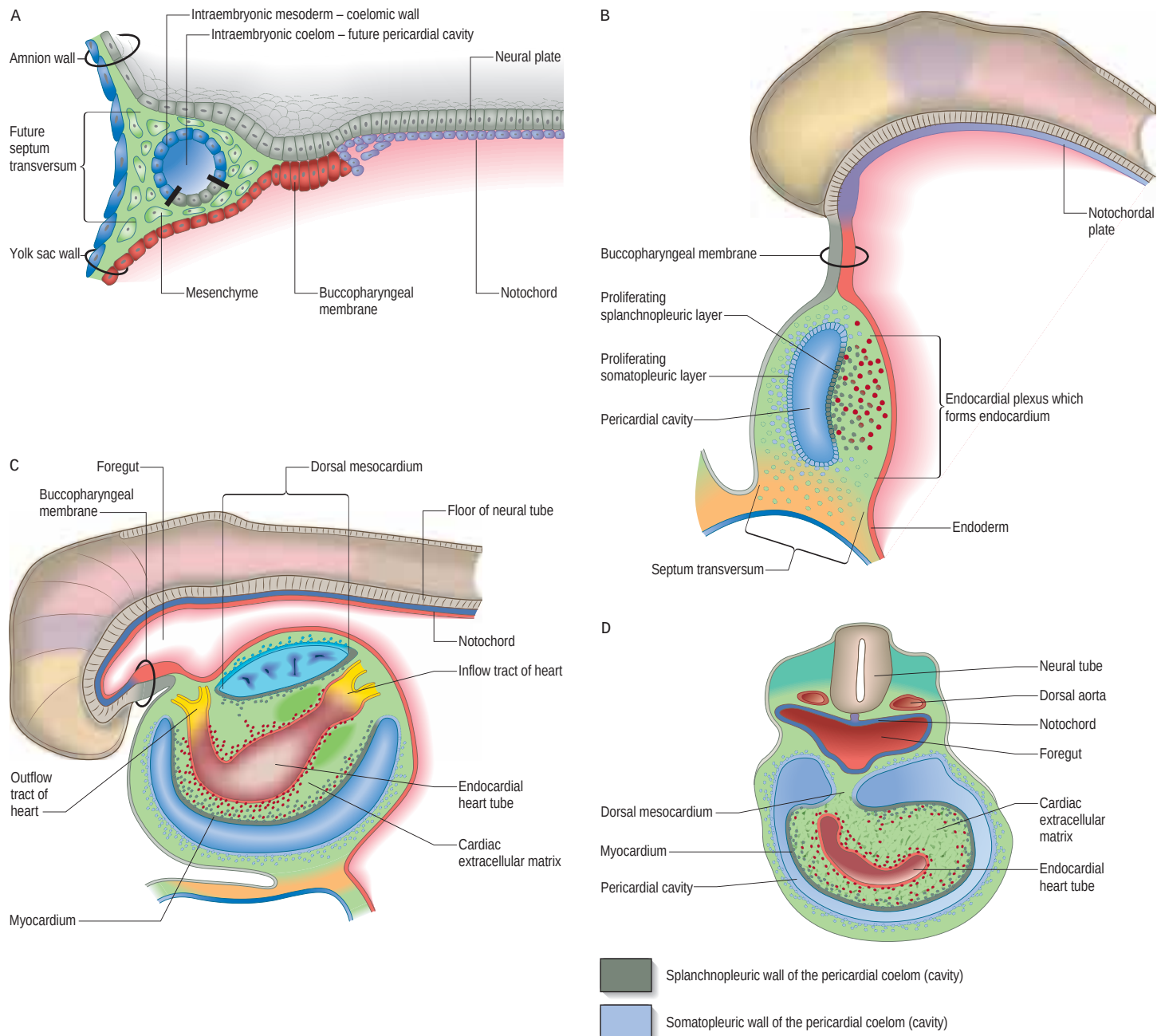


Fig. 52.7 **A**, A median section through part of the embryonic disc to show the position of the future pericardial cavity before head folding. **B**, A median section through the cranial end of a human embryo during early head folding, showing its reversal effect on the position of the pericardium. **C**, A median section through the cranial end of a human embryo after head folding, showing the pericardial cavity and endothelial heart tube now ventral to the foregut. **D**, A horizontal section through the pericardium and endothelial heart tube shown in C. The position of the dorsal mesocardium, a source of secondary heart field mesenchyme, is indicated.

in the spatio-temporal regulation of the division of the outflow tract into the aortic and pulmonary pathways, and the development of the muscular subpulmonary infundibulum.

The epicardium, sometimes included in descriptions of the myocardium as 'epimyocardium', is not present at the early stages of heart development.

Endocardial cushions

The extracellular matrix of the heart, historically termed cardiac jelly, promotes occlusion of the endocardial tubular lumen during myocardial contraction, thus providing mechanical assistance for the generation of the flow of blood. It also acts as a site for the deposition of inductive factors from the myocardial cells, which, in turn, modify the differentiation of specific endocardial cells. It has been called a gelatinoreticulum, a myoepicardial reticulum (see Fig. 52.7C–D) and, more recently, the myocardial basement membrane. Here, the term cardiac extracellular matrix will be used. It is composed of hyaluronic acid, hyaluronidase and fibronectin, among other things. Inductive signals originating from the myocardial cells cause a subset of endocardial cells

lining the atrioventricular canal and the proximal outflow tract to transform into mesenchyme (cardiac mesenchyme); the endocardial cells in other regions of the heart tube, such as those in the ventricle, do not undergo such a transition. When activated by myocardial inductive factors, the endocardial cells lose their cell-to-cell associations, and show decreased expression of neural cell adhesion molecule, and increased expression of substrate adhesion molecules such as chondroitin sulphate and fibronectin. They undergo rearrangement of their cytoskeleton necessary for migration, and they express type I procollagen. Uniquely, they retain their expression of endothelial markers. This epithelial to mesenchymal transition may, perhaps, be the only example of a mesenchymal population that is derived from an endothelial lineage.

Formation of cardiac mesenchymal cells at the atrioventricular canal and the proximal myocardial outflow tract is followed by their migration into the cardiac extracellular matrix. These cells proliferate between the endocardium and myocardium, and, with local accumulation of extracellular matrix molecules, produce protrusions, termed endocardial or cardiac cushions, which bulge into the primary heart tube and initially provide the valvular mechanisms required in the atrioventricular canal and outflow tract. Their position corresponds to

the future positions of the definitive cardiac valves. In the distal part of the outflow tract, which initially has myocardial walls, cells that are derived from the neural crest subsequently make significant contributions to the mesenchyme of the endocardial cushions. Although proper migration of these cells from the neural crest is crucial for normal development of the outflow tract and formation of the leaflets and sinuses of the arterial valves, their function is largely obscure. They are no longer found in the leaflets of the arterial valves in the formed heart, or in the muscular subpulmonary infundibulum, which is also derived from the outflow cushions. The endocardial cushions themselves ultimately fuse, forming a wedge of mesenchyme that serves to guide the union of the internal muscular septal structures. At their time of fusion, the atrioventricular endocardial cushions are large, relative to the size of the atrioventricular orifices; they will provide the scaffold for formation of the leaflets of the tricuspid and mitral valves.

Cardiac myocytes – contraction, conduction and automaticity

Cardiac myocytes share a number of characteristic features that distinguish them from other cells. All cardiac myocytes have sarcomeres and a sarcoplasmic reticulum; in principle, they share the capacity for producing an intrinsic cycle of electrical activity resulting in contraction. This phenomenon is called automaticity, or pacemaker activity. An absolute requirement for effective pacemaking is poor electrical coupling of the cells, which also implies slow conduction. It allows the cells to build up sufficient electrical charge, which is then propagated through the surrounding myocardium. Thus ‘a small node can drive a large heart’.

Varying degrees of differentiation are seen in early populations of cardiac myocytes, which can be categorized as forming working, nodal, conducting and primary myocardium (Table 52.1). Cells of the atrial and ventricular working myocardium display virtually no automaticity, but are well coupled and have well-developed sarcomeres and sarcoplasmic reticular structures. The development of the synchronously (fast) contracting working myocardium requires fast conduction of the depolarizing impulse, and so the cells possess well-developed gap junctions. In marked contrast, the cells forming the nodes of the cardiac conduction system have the opposite phenotype, and resemble the myocytes that are found initially in the primary heart tube. The cells of the putative atrioventricular and peripheral ventricular conduction system have an ambiguous phenotype: the cells are well coupled, thus allowing fast conduction of the depolarizing impulse, but otherwise retain an embryonic phenotype. The division does not imply that the cells belonging to one group are identical, but rather that they share distinguishing features developed to variable degrees.

Automaticity and slow conduction are features of the myocardium of the primary heart tube (see Video 52.1). The slow conduction of the depolarizing impulse over the heart tube causes a peristaltic wave of contraction, by means of which the blood is pushed towards the arterial pole. During the waves of contraction, the accumulated cardiac extracellular matrix closes the endocardial tube and prevents the backward flow of blood. Unambiguous morphological markers are often lacking in the early embryonic heart, which means that it is not possible to distinguish the various cells phenotypically; automaticity and the speed of conduction have, therefore, become important functional parameters by which to describe the development of the different parts of the heart.

HEART TUBE

During the process of embryonic folding, the midline splanchnopleuric coelomic epithelium, which is derived from the primary heart field (cardiac crescent), becomes positioned ventral to the foregut. The coelomic epithelium is highly proliferative and gives rise to mesenchymal populations. The early cardiac myocytes arise from the coelomic wall adjacent to the endoderm of the foregut (see Fig. 52.7). The cells

undergo restriction, differentiate and leave the cell cycle. After folding, the second heart field contributes cells to the heart. It has been suggested that the secondary heart field may contribute cells only to those cardiac components that are required for the pulmonary circulation: namely, the right ventricle and outflow tract at the arterial pole, and the atrial septum and the dorsal atrial wall at the venous pole. To extend this suggestion, the original primary heart-forming field would give rise to those components that are required for the systemic circulation: namely, the systemic venous sinus and its tributaries, the initial atrium, the left ventricle and the arterial conus, as seen in the outflow tract of primitive fishes, although this latter structure has no homologue in mammalian hearts.

During the process of folding, the pericardial cavity and, concomitantly, the myocardium gradually extend around the forming endocardial tube, leaving the dorsal mesocardium, a transient connection that is analogous to the mesentery of the intestines (see Fig. 52.7C–D). The persisting stalk of the dorsal mesocardium connects the venous pole of the heart with the splanchnopleuric mesenchyme around the developing lung buds and with the septum transversum mesenchyme, which will give rise to the liver. The dorsal mesocardium is the site of early mediastinal mesenchyme production. It disappears as a mesenteric entity during the third week of development, when the embryo has from 4 to 12 somites; at the same time, the endocardial heart tube becomes entirely surrounded by the myocardium and enclosed within the pericardial cavity. The breakdown of the dorsal mesocardium establishes a passage across the pericardial cavity, from side to side and dorsal to the heart, which persists as the transverse sinus of the pericardium. After folding and rupture of the dorsal mesocardium, the second heart field contributes cells to both the arterial and the venous poles of the heart.

At this stage, the heart is bilaterally symmetrical and has the form of an inverted ‘Y’. The legs of the Y are positioned caudal to each side of the cranial intestinal portal, where they form the venous pole of the heart (Fig. 52.8A–B). The stem of the Y connects cranially with the great arteries, forming the arterial pole of the heart. From the outset, both poles are connected with the systemic vessels running from the embryo, the yolk sac and the placenta. The pulmonary veins form later, during the fifth week of development, the initial venous primordium becoming evident in the mesenchyme derived from the dorsal mesocardium caudally. Also at this site, mediastinal myocardium continues to be added to the heart, forming the smooth dorsal wall of the left atrium, and providing the site of formation of the primary atrial septum.

Recent genetic fate map studies have shown that the primary heart tube contains cell lines that will contribute to only two compartments. The myocardium of the stem of the inverted Y contains only precursor cells of the left ventricle, and both legs of the inverted Y contain precursor cells of the atrioventricular canal and parts of both atrial chambers. The atrioventricular canal, in turn, provides the precursors for the left ventricular free wall, whereas the cells of the original embryonic left ventricle largely end up in the ventricular septum. The embryonic outflow tract contributes to the right ventricle. There is no evidence to support the previous, traditional, concept of heart development, i.e. that all the putative cardiac segments are already present in the straight heart tube and develop to contribute to the various parts of the definitive cardiac chambers. This hypothesis has now been disproved by elegant experimental studies that have shown that the definitive cardiac chambers are formed by local differentiation and expansion, or ‘ballooning’, of the myocardial walls of the primary heart tube.

Looping of the heart tube

Cranial and caudal recruitment of myocardium and endocardium produces considerable lengthening of the primary heart tube. Recent studies of the mechanism of proliferation in the avian, mouse and human heart have shown that cardiac myocytes stop dividing after overt differentiation. The tube continues to elongate, largely as a result of the recruitment of additional myocytes to the tube, as shown by studies of both morphology and molecular lineages, and, to a lesser extent, because of an increase in the size of the myocytes that form the walls of the tube. The lengthening heart tube bends ventrally and rightwards, concomitant with the breakdown of the dorsal mesocardium. The bend is called the ventricular loop, since the left ventricle subsequently balloons and expands at its outer curvature, which is the original ventral side of the straight tube. The inner curvature of this loop was originally its dorsal side.

It is often stated that looping of the tube is the first visual evidence of asymmetry in the embryo, although careful examination reveals that the atrioventricular canal has become asymmetric prior to the start of

Table 52.1 Basic phenotypes of cardiac myocytes

Feature	Type of myocardium			
	‘Primary’	‘Nodal’	‘Conducting’	‘Working’
Automaticity	High	High	High	Low
Conduction velocity	Low	Low	High	High
Contractility	Low	Low	Low	High
Sarcoplasmic reticulum	Low	Low	Low	High

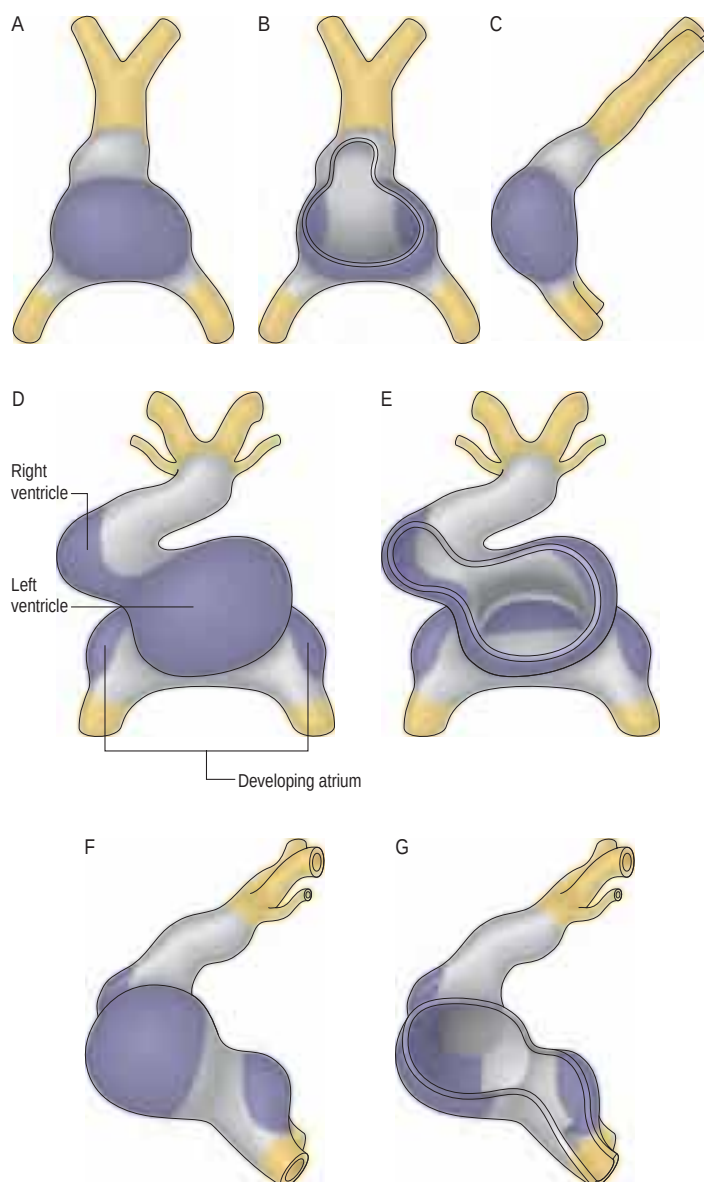


Fig. 52.8 Local expansion of the myocardial wall of the early heart tube and heart tube looping. **A**, A ventral view. **B**, The ventral wall has been removed to show the inner aspect of the early heart tube. **C**, A left lateral view. The early heart tube is shaped like an inverted 'Y'. The legs of the Y pass caudal to each side of the cranial intestinal portal. In the early heart tube, the myocardium of the stem of the Y (purple) contains only precursor cells for the left ventricle. The remaining myocardium (grey) contains only precursor cells for the atrioventricular canal and atria. The precursor left ventricular cells expand first. **D–G**, The later heart tube, showing looping. **D**, A ventral view. **E**, The ventral wall has been removed to show the inner aspect of the folding heart tube. **F**, A left lateral view. **G**, The lateral wall has been removed to show the inner aspect of the heart from the left. In **D** and **E**, the expansion of the right ventricle can be seen and the heart tube has elongated to the right. The developing atrium can be seen dorsolaterally in **D–G**.

looping. Although the sense of laterality of the developing organs of the body, including the atrial appendages, develops during gastrulation, the pathway of signalling that governs rightward looping of the heart tube remains unknown. However, it is now well established that signalling pathways, including *Pitx2*, *nodal*, *lefty* and *cited-2*, determine the formation of the morphologically left-sided or right-sided features seen in organs such as the lungs, bronchial tree, liver and spleen, and atrial appendages.

By the time the primary heart tube has formed its ventricular loop, it is possible to distinguish the atrial and ventricular components of the developing heart (Figs 52.8D–G, 52.9), and to recognize an outflow tract connected with the aortic sac and developing pharyngeal arch arteries. The cells that, at this stage, make up the outflow tract, will, at later stages, be found within the right ventricle. At the proximal side of the outflow tract, cells are recruited to the developing right ventricle and, contemporaneously, new cells are recruited to the outflow tract from the second heart-forming field. The atrial and ventricular compo-

nents are separated one from another by the atrioventricular canal, which, at this stage, has significant length. The systemic venous tributaries drain directly to the atrial cavity.

The sinus venosus with right and left sinus horns, which forms at the confluence of the systemic venous tributaries, is a prominent structure in lower vertebrates. In mammals, it does not become recognizable as a morphological entity until the systemic veins have excavated themselves from the mesenchyme of the body and been surrounded by myocardium. It then becomes incorporated into the pericardial cavity so that the sinus horns become part of the heart. All of these processes occur up to the twenty-fifth day of development in humans; only after this stage is it possible to consider development of the definitive atrial and ventricular chambers.

INFLOW TRACT

The working myocardium of the atrium differentiates locally at the dorsolateral sides of the heart tube (see Fig. 52.8D–G). The developing atrium then expands enormously, in dorsal, lateral and, most prominently, cranial directions. The cranial expansion is seen as pouches, which become the left and right atrial appendages (see Fig. 52.9A–B). The floor of the atrium, including the sinus venosus, and the atrioventricular canal are made of primary myocardium. Less cellular proliferation occurs in these parts than in the expanding atrial chambers; the primary myocardium marks the inflow to, and the outflow from, the initial atrial chambers as 'rings', which are also associated with the formation of the conduction system. The ring at the inflow defines the sinuatrial junction, whilst the atrioventricular canal, which forms the atrial outlet during development, will eventually be incorporated into the definitive right and left atrial chambers as the atrial vestibules. Although it is possible to recognize the forming left and right atrial appendages at this stage, the right being more extensive than the left, the atrium itself has a single cavity and there is no evidence of septum formation. The myocardium of the sinus venosus and the newly forming mediastinal myocardium are smooth-walled, whereas the myocardium of the appendages shows ridges, the pectinate muscles, on the inner surface. The formation of the different appendages is under control of the *Pitx2* signalling pathway. The myocardium of the appendages has a chamber phenotype, or is working myocardium; it expresses atrial natriuretic factor and connexin40, among other markers.

Right atrium

The further development of the right atrium is characterized by the incorporation of the sinus venosus into the right part of the primary atrium. This process is under control of the T-box transcription factor *Tbx18*. At about 4 weeks of development, the sinuatrial junction of the looping primary heart tube is positioned symmetrically in the midline (Fig. 52.10; see also Figs 13.1, 60.9). The left and right common cardinal veins drain directly into the cavity of the primary atrium. The atrial myocardium extends to the margins of the pericardial cavity and, strictly speaking, the sinus venosus is not yet formed because the systemic venous tributaries are embedded within the mesenchyme of the septum transversum.

During subsequent development, the pericardial cavity expands to enclose the terminal segments of the systemic venous tributaries; at the same time, their walls differentiate as myocardium. They can now be termed the left and right horns of the sinus venosus; each horn receives the union of the corresponding umbilical vein, vitelline radicles and common cardinal vein (see Figs 52.10, 60.9). Concomitantly, the constriction between the left horn and the atrium becomes more pronounced. As the dorsal wall of the left atrium is formed from additions of mediastinal myocardium, the left horn becomes incorporated into the developing left atrioventricular junction, its orifice draining to the newly formed right atrium. At the same time, the left-sided venous tributaries diminish in size; the left common cardinal vein forms the oblique vein of the left atrium, and the left sinus horn forms the coronary sinus (see Fig. 52.10C), maintaining its own myocardial wall as it becomes incorporated into the atrioventricular junction.

The right sinus horn increases rapidly in size with growth of the liver (see Figs 60.9–60.10). The vitello-umbilical blood flow enters the right horn through a wide but short hepatocardiac channel, which becomes the cranial end of the inferior vena cava. The right horn also receives the right common cardinal vein, draining the blood from the right side of the body. Later, when transverse connections are established between the cardinal veins, the blood from the left side of the body also reaches the heart via the veins draining the right side (see Fig. 13.4). As these

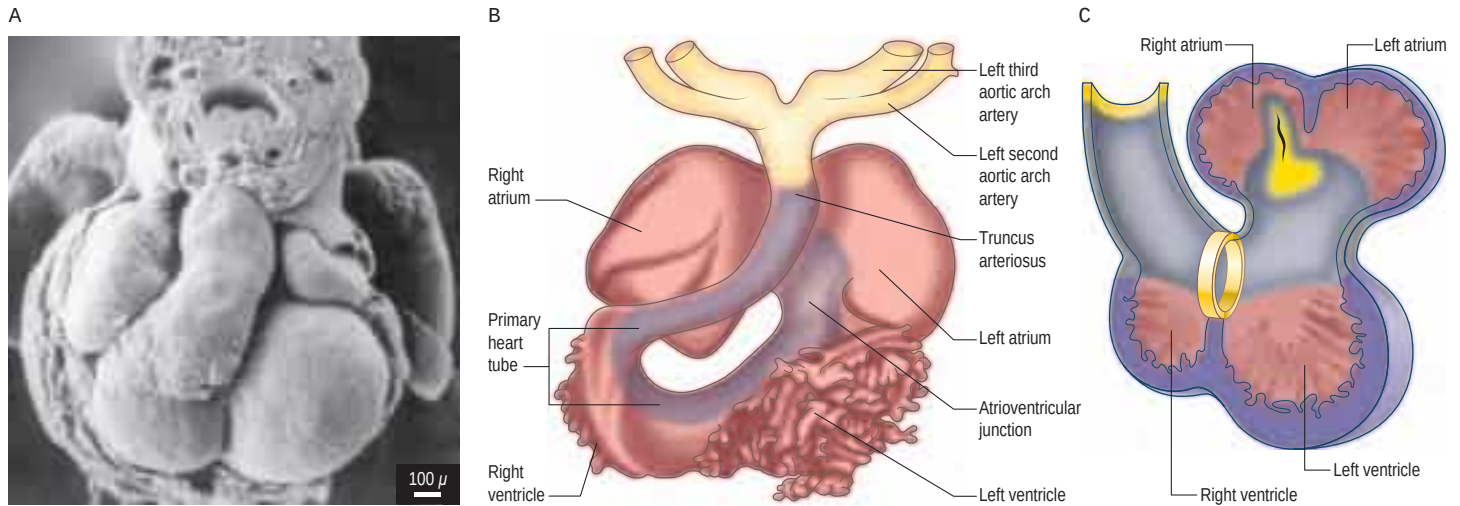


Fig. 52.9 **A**, A scanning electron micrograph of an early developing human heart, showing the external appearance of the myocardium of the right and left ventricles and the right and left atrial appendages. The truncus arteriosus passes between the atria. **B**, A reconstruction of the heart tube at stage 13, showing the endocardium. In the left and right ventricles, the endocardium interdigitates with the myocardium and appears trabeculated; elsewhere, it is separated from the myocardium by the cardiac extracellular matrix. **C**, The heart at stage 13 with the truncus arteriosus reflected to the right. The opening of the sinus venosus into the right atrium can be seen (yellow). The region of the primary heart tube is indicated by a yellow interventricular ring. (A, With permission from Oostra RJ, Steding G, Lamers WH, Moorman AFM. Steding's and Virágh's Scanning Electron Microscopy Atlas of the Developing Human Heart. New York: Springer-Verlag, 2006. B, Modified from O'Rahilly and Müller. Developmental Stages in Human Embryos 1987 Carnegie Institution of Washington. Pub 637.)

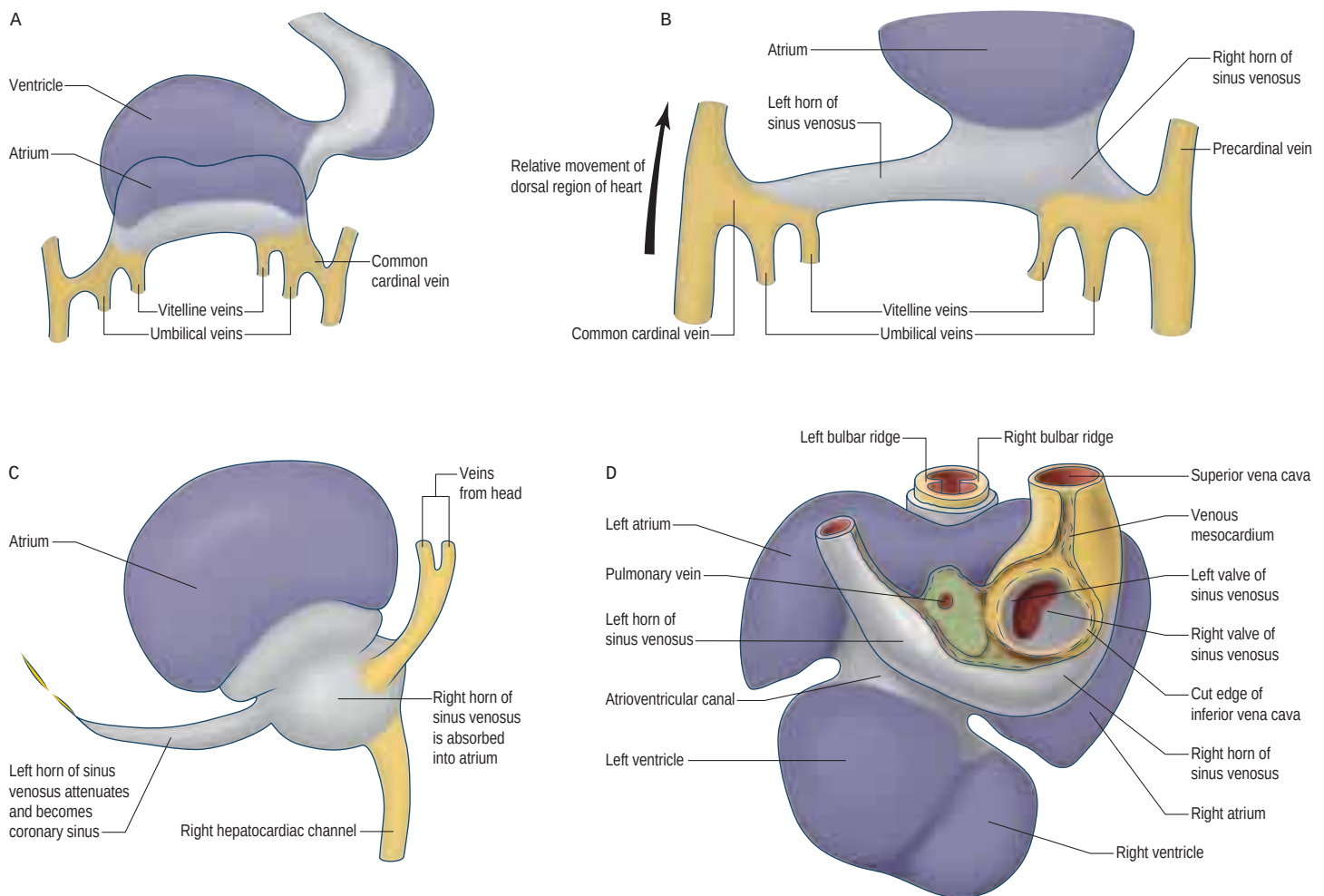


Fig. 52.10 The changes to the sinus venosus. **A**, A dorsal view of the early heart tube shown in Figure 52.8D. **B**, Changes to the circulation bring the venous circulation to the right, causing enlargement of the right horn of the sinus venosus and atrium, and attenuation of the left horn of the sinus venosus. **C**, The right horn of the sinus venosus becomes absorbed into the atrium. The left horn of the sinus venosus becomes the coronary sinus. **D**, A dorsal view of the embryonic heart, showing the relative changes to the sinus venosus.

changes take place, the right sinus horn, including the proximal parts of the superior and inferior cardinal veins, becomes incorporated into the right atrium, forming the smooth-walled systemic venous sinus, also known as the sinus venarum.

The right sinus horn opens into the right atrium through its dorsal and caudal walls (see Figs 52.10, 52.14). The sinuatrial orifice becomes elongated and slit-like, guarded by two muscular folds: the left and right sinuatrial (venous) valves. These two valves meet cranially and become continuous with a fold that projects from the atrial roof: the septum spurium. The valves also meet caudally, and merge with the inferior atrioventricular cushion. With ongoing development, the cranial part of the right sinuatrial valve loses its fold-like form, but its position is indicated in the adult heart by the site of the crista terminalis of the right atrium. Its caudal part forms the valve of the coronary sinus, also known as the Thebesian valve, and most of the valve of the inferior vena cava (Eustachian valve). The union of the two valvular remnants then passes through the tissue that separates the orifice of the coronary sinus from the fossa ovalis. This area is known as the 'sinus septum' but, in reality, this 'septum' is no more than a muscular fold in the dorsal wall of the right atrium. The continuation of the venous valves persists as the tendon of Todaro, an important landmark for the location of the atrioventricular node in the definitive heart. The left venous valve blends with the right side of the atrial septum; there is usually no trace of it in the postnatal heart.

Left atrium

Following the formation of the primary atrium and the left atrial appendage, the left atrium takes shape by the formation and incorporation of mediastinal myocardium. While the tributaries of the sinus venosus approach the atrium caudally, the differentiating pulmonary veins gain their entrance to the atrial cavity through the dorsal mesocardium (see Fig. 52.7). The topographical relationships seen in the postnatal heart are, thus, established as soon as differentiation of the pulmonary venous portal occurs.

Early in the development of the atrium, the pulmonary vein develops as a solitary channel from angiogenic cells derived from the dorsal mesocardium, and establishes continuity with the vascular plexus formed in the mediastinal mesenchyme around the developing lung buds. The solitary pulmonary vein opens into the caudo-dorsal wall of the atrium adjacent to the developing atrioventricular junction, its atrial orifice being flanked by two prominent ridges. The primary atrial septum develops from the right ridge, following incorporation of the left sinus horn into the right side of the primary atrium, thus confining the pulmonary venous orifice to the developing left atrium (see Fig. 52.14). The pulmonary vein initially branches within the dorsal mediastinal mesenchyme, its tributaries draining blood from the developing lung. With continuing development, the walls of the venous channels become surrounded by myocardium, this process occurring to the level of the second bifurcation. The veins then expand and are incorporated into the roof of the left atrium, eventually forming the greater part of its cavity. However, all four pulmonary veins do not achieve their separate opening into the atrial roof until well after the completion of atrial septation, and probably not until the tenth or eleventh week of development. Variations in the precise pattern of pulmonary venous drainage are quite common. With these changes, the left half of the primary atrium becomes progressively restricted to the mature appendage. The myocardial sleeves surrounding the pulmonary orifices taper off and become intermingled with fibrous tissue. In later life, it is likely that this intermingling of myocardial and fibrous tissues forms the substrate for some forms of atrial fibrillation. The opening of the solitary pulmonary vein to the left of the right pulmonary ridge is a prerequisite for atrial septation.

VENTRICLES

The ventricles develop at the ventral side of the looping primary heart tube during the fourth week of development (see Figs 52.8–52.9). The left ventricle develops from the stem of the Y-shaped heart; the right ventricle develops later, downstream relative to the left ventricle, when more myocardium has been added to the cardiac tube. As a result of the looping of the heart tube, the right ventricle is positioned at the right of the left ventricle, which is a prerequisite for the appropriate connection with the expanding atrial component of the heart. Unlike the atrial chambers, the morphological differences between the right and left ventricles are not part of the general asymmetry between the right- and left-sided organs of the body, but rather are under control of

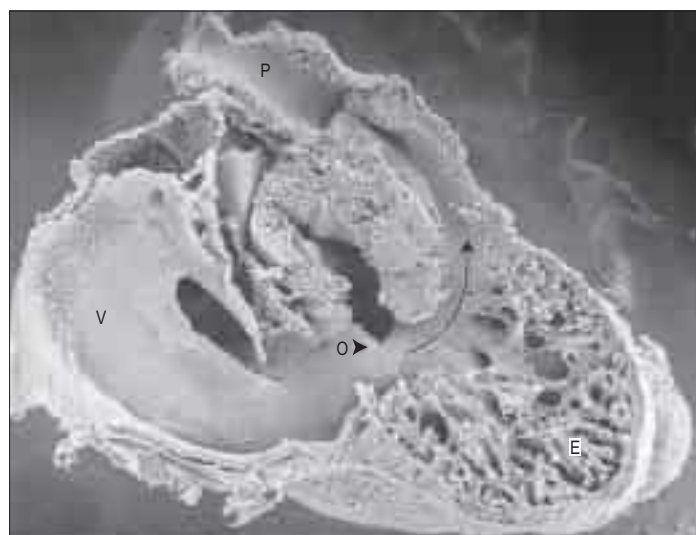


Fig. 52.11 A scanning electron micrograph of a human heart at 42 days' gestation. The right lateral wall of the right atrium, the right ventricle and the outflow tract have been removed. The valve of the sinus venosus (V) is prominent. There is no septal cusp of the tricuspid valve guarding the inlet portion of the right ventricle. The parietal cusp has been removed. The atrioventricular cushions are fused. Note the adjacent right lateral tubercles (O). A small interventricular foramen (foramen secundum, p. 915) is indicated (arrowhead). There is a well-developed right ventricular outflow tract (curved arrow). Other abbreviations: E, embryonic trabeculae; P, pulmonary trunk.

the signalling pathways that determine caudocranial differentiation. Retinoic acid and its downstream transcription factor, Tbx5, play a crucial role in this process.

The myocardium at the inner curvature, the original dorsal side of the cardiac tube, remains smooth-walled and maintains its molecular phenotype, whereas the myocardium at the outer curvature of the myocardial tube displays trabeculations in the fifth week of development; the original primary heart tube is represented by an interventricular ring (see Fig. 52.9C). By stage 17, the trabeculations have achieved a typical spatial orientation, giving a sponge-like appearance to the internal aspect of both ventricles (Fig. 52.11). The definitive trabeculations, coarse in the right ventricle but much finer in the left, are first observed around the fortieth day of gestation; they appear initially in the walls of both ventricles at the level of the atrioventricular junction and develop towards the apex of the heart. By the time the fetus is 10 weeks old, the trabeculations are much sparser, and are confined to the apical regions. This process of remodelling is accomplished without the intervention of macrophages or inflammatory cells in the immediate interstitium. The ventricular myocardium, encompassing the trabeculations and exterior wall, possesses a chamber phenotype, the myocytes expressing, among other proteins, the fast-conducting gap-junctional proteins connexin 40 (Cx40) and 43, and atrial natriuretic peptide. This myocardium stops proliferating and differentiates into the fast-conducting peripheral ventricular conduction system, whereas the outer layer becomes highly proliferative and forms the compact layer of the ventricular wall, no longer expressing Cx40 and atrial natriuretic peptide, but still expressing Cx43.

OUTFLOW TRACT

Initially, the outflow tract possesses myocardial walls up to the pericardial reflections. It displays a characteristic bend, which has a proximal part arising from the developing right ventricle, and a distal part that becomes continuous with the aortic sac beyond the pericardial reflections. Although this bend is a conspicuous morphological feature, it is highly unlikely that the cells to either side of it retain their position during subsequent development. Instead, it seems probable that cells are recruited from the second heart-forming field, and pass through the outflow tract to contribute to the definitive right ventricle. Subsequently, still more cells are recruited from pharyngeal mesenchyme, passing proximal to the pericardial reflections and forming the intrapericardial components of the arterial trunks. There remains much disagreement as to how best to describe these morphological parts of the outflow tract. Some authors have labelled the proximal myocardial part the conus, calling the distal part the truncus, and the intrapericardial portion of the outflow tract the aortic sac. Others have called the entire

myocardial outflow tract the bulbus, since the myocardial outflow tract of primitive vertebrates, called the conus arteriosus in primitive fish and the bulbus cordis in amphibia, is presumed to develop into the right ventricle in higher vertebrates. Because the terms conus and truncus have been inconsistently used in literature, we prefer simply to describe proximal and distal parts of the myocardial outflow tract, and to describe, when it appears, an intrapericardial arterial portion of the outflow tract, this part interposing between the muscular component and the ventral aorta, the distal junction occurring at the pericardial reflections.

Given that different authors use different terminologies, it is not surprising that formation of the outflow tract remains one of the most confusing areas of cardiac development. The recent finding that the stem of the heart tube contains precursor cells exclusively for the left ventricle makes it essential to adopt descriptions that follow dynamic events, rather than continuing to use static names. Thus, as soon as the primary heart tube bends to produce its ventral ventricular expansion, by definition, the developing ventricle achieves inlet and outlet components, even though these structures are not the definitive ventricular inlets and outlets. Cells that, initially, are found in the ventricular outflow tract subsequently become cells of the right ventricle. Furthermore, working myocytes can easily be distinguished from the primary myocytes of the straight heart tube. The primary myocardium is smooth-walled, whereas the developing myocardium of the ventricles is trabeculated, and expresses specific markers such as atrial natriuretic peptide and Cx40 and Cx43. Thus, that part of the primary heart tube downstream of the developing ventricular loop is termed the outflow tract, irrespective of the fact that, eventually, it will largely become incorporated within the definitive right ventricle (see Figs 52.8–52.9). Moreover, within the inner curvature of the ventricular loop, the walls of the outflow tract and atrioventricular canal fade into one another without a clear boundary. Indeed, ventricular working myocardium never develops at the inner curvature. It is within this ventricular part of the primary heart tube that the cushions of the atrioventricular canal and the outflow tract must achieve appropriate connections with the muscular ventricular septum in order to divide this part of the tube into left and right compartments. The cavity of the tube within which these events take place has been called the ‘interventricular foramen’. This is incorrect, since it is, rather, the region between adjacent parts of the primary heart tube. It is, none the less, the remodelling of this inner curvature that sets the scene for the completion of cardiac septation.

CARDIAC FUNCTION AND THE CONDUCTION SYSTEM

Although only mammalian hearts have a well-defined conduction system, an essentially similar electrocardiogram can be recorded from animals as diverse as fish and humans, indicating that the electrical connections between the cardiac components, along with the underlying gene-regulatory networks, have been conserved during vertebrate evolution. Indeed, the development of the conduction system of the heart is inextricably associated with the development of the basic building plan for the heart. The conduction system is best defined as the system that initiates and conducts the sinus impulse. In mammals, it encompasses the sinuatrial node, the atrioventricular node, the atrioventricular bundle, the bundle branches and their terminal ramifications. The nodes and the atrioventricular bundle can be considered the central conduction system, while the bundle branches and their ramifications represent the peripheral ventricular conduction system. The sinus impulse is generated in the sinus node; from here, it is rapidly propagated through the atrial myocardium towards the atrioventricular node, where propagation is delayed. After an essential period of atrioventricular delay, the impulse travels rapidly through the atrioventricular conduction axis and the peripheral ramifications to reach the ventricular myocardium, which is then rapidly depolarized. Following repolarization, the sequence recurs in endless fashion throughout the life of the individual. The term ‘conduction system’, as opposed to the working myocardium of the chambers, may suggest that the working myocardium does not conduct, while the conduction system conducts rapidly. In order to produce powerful synchronous contractions, the working myocardium of the chambers must also, of necessity, conduct rapidly, whereas to act as a pacemaker, the cells of the nodes need to be poorly coupled, and hence display slow conduction.

The vertebrate heart is myogenic, implying that all cardiomyocytes have the capacity to generate and conduct the depolarizing impulse. Because cardiac myocytes are electrically coupled, the cells with the highest pacemaking activity take the lead. In the formed normal heart, the leading pacemaker is always within the sinus node. In embryonic

hearts, the leading pacemaker is always found at the venous pole (Fig. 52.12). Thus, it is the most recently recruited cells at the venous pole that always have the highest pacemaking activity. All regions of the early embryonic heart tube possess poorly coupled cells and display intrinsic automaticity, thus permitting slow propagation of the depolarizing impulses along the cardiac tube, and producing matching peristaltic waves of contraction that push the blood in an antegrade direction. The myocytes of a developing heart do not have well-developed sarcomeric structures, and have a poorly developed sarcoplasmic reticulum. This tissue has been called primary myocardium, distinguishing it from the rapidly conducting and well-developed working myocardium of the atrial and ventricular chambers.

As development proceeds, the ventricular chambers develop in the cranial part of the heart tube, by ventral expansion, and the atrial chambers in the caudal part by dorsolateral expansion (see Fig. 52.8). An adult type of electrocardiogram can be recorded from such hearts, showing rapid atrial depolarization, a period of atrioventricular delay, and rapid ventricular depolarization. The electrocardiographic tracings reflect the development of fast-conducting components within a slowly conducting heart tube, i.e. the development of a morphological conduction system. The newly developed working myocardium expresses atrial natriuretic factor, along with the gap junctional proteins Cx40 and Cx43, which permit fast conduction.

Like the early peristaltic heart, flow through the developing chambered heart remains directional because the dominant pacemaking activity is still at the inlet of the heart (see Fig. 52.12). Peristaltic hearts do not need valves, whereas chambered hearts require the presence of one-way valves at the inlets to, and the outlets from, the chambers. The slowly conducting atrioventricular canal, interposed between the atrium and the ventricle, and the outflow tract positioned between the ventricle and the great arteries, are the parts that contain the endocardial cushions. These are able to function as sphincteric valves, as a result of their prolonged duration of contraction. These components retain their function until the definitive leaflets of the one-way valves have been sculpted from the cushions. The primary myocardium of the outflow tract does not regress until the arterial valves have been formed, and does not disappear in its entirety until around the twelfth week of human development. The primary myocardium of the atrioventricular canal is eventually incorporated into the atrial vestibules at the same time as the formation of the atrioventricular valves, becoming sequestered on the atrial side of the atrioventricular junctions between 6 and 12 weeks of development. An important part of the canal persists dorsally, where it differentiates into the slowly conducting atrioventricular node (see Fig. 52.12).

Considerable progress has been made over recent years in deciphering the pathways involved in establishing the building plan of the heart (Fig. 52.13). The combined action of the transcription factors Tbx5 and Nkx2-5 is required for the formation of the myocardium of the atrial and ventricular chambers. Tbx5 is expressed in a gradient over the heart tube, decreasing in concentration from caudal to cranial, and may impose positional information. However, its pattern cannot explain the localized formation of the chambers and the conduction system. Localized expression of the transcriptional repressors Tbx2 and Tbx3 in the inflow tract, the floor of the atrium, the atrioventricular canal, the inner curvature of the ventricular region, and the outflow tract prevents the differentiation of primary into working myocardium. Tbx2 and Tbx3 effectively compete for binding to the promoters of the chamber-specific genes, such as atrial natriuretic factor, Cx40 and Cx43. The remaining primary myocardium of these regions initially induces the endocardium to undergo epithelial to mesenchymal transition, by which process the cardiac cushions become filled with mesenchyme. The myocardium also participates in the alignment of the atrial and ventricular chambers (which is essential for proper cardiac septation, and is guided by the fusing cardiac cushions); it forms the conduction system, including the sinus and atrioventricular nodes, the atrioventricular bundle, the floor of the developing right atrium (the internodal region), and the entirety of the atrioventricular canal. In the early chamber-forming heart of the mouse, Tbx3 is expressed in the atrioventricular canal, the floor of the atrium and around the orifices of the systemic venous tributaries, but not in the dorsal mesocardium surrounding the entrance of the pulmonary vein. During subsequent development, the domain of expression of Tbx3 expands from the atrioventricular canal to form a crescent on the crest of the ventricular septum. The dorsal aspect of this crescent develops into the atrioventricular bundle, while the atrioventricular node is formed at the dorsal junction with the atrioventricular canal. The ventral parts subsequently disappear during development.

The fact that the ventricular conducting system originates from a single primary ring provides a solid base for understanding the

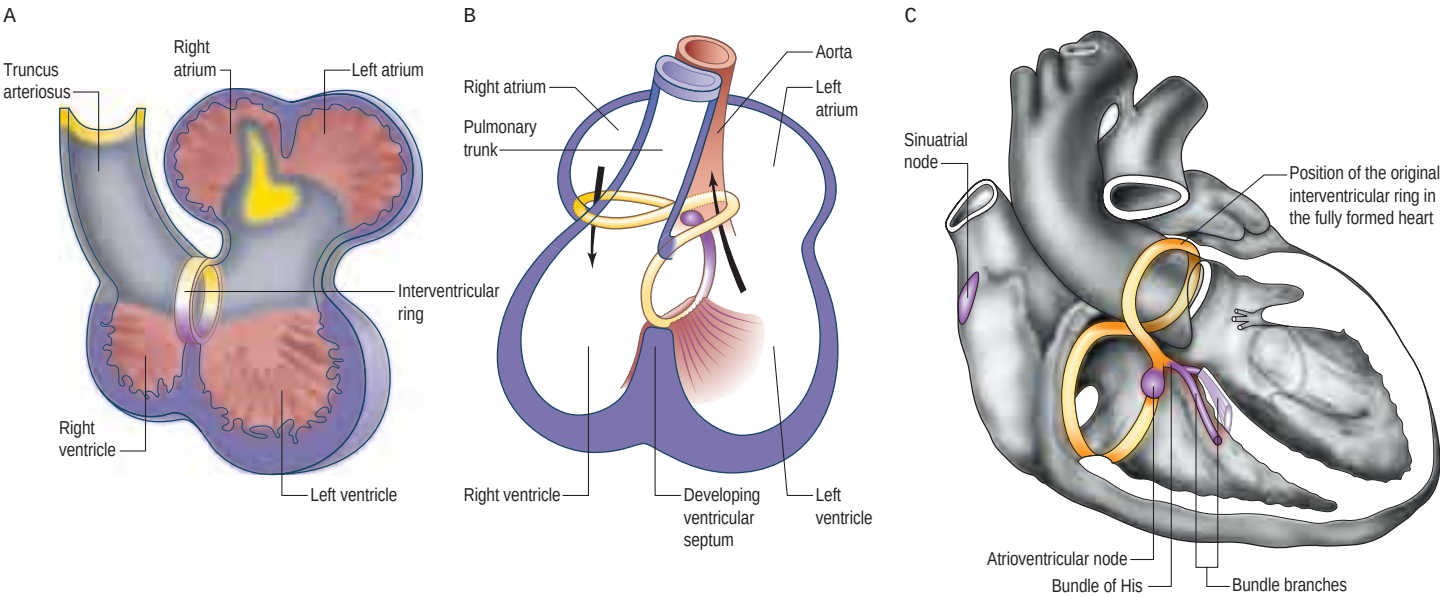
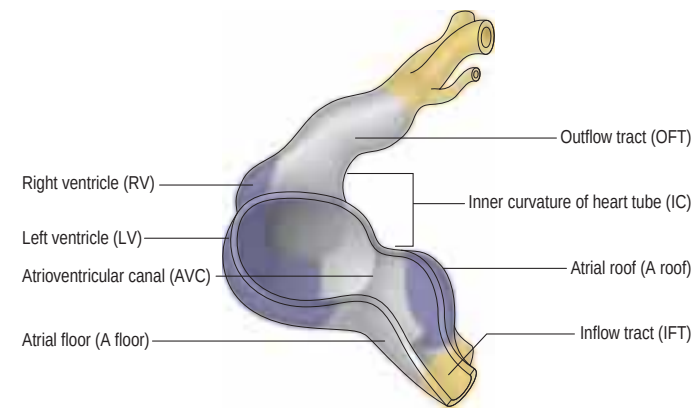


Fig. 52.12 Development of the conduction system of the heart. **A**, At stage 13 with the truncus arteriosus reflected to the right, the interventricular ring marks the region of the primary heart tube. The purple part of the interventricular ring will give rise to the atrioventricular bundle and bundle branches, whereas the yellow part of the interventricular ring will not participate in the formation of the adult conducting system. **B**, The truncus arteriosus, now divided into the pulmonary trunk and aorta, is presented between the right and left atria, which expand lateral to it. The right atrium is now positioned entirely above the right ventricle, and the outflow tract has expanded to the left; the left ventricle has access to the aorta. **C**, The position of the original primary heart tube ring indicated in the formed heart. The pulmonary trunk has been removed.



	IFT	A roof	A floor	AVC	IC	LV	RV	OFT
Tbx1	–	–	–	–	–	–	–	+
Tbx2	+	–	+	+	+	–	–	+
Tbx3	–	–	+	+	+	–	–	–
Tbx5	+	+	+	+	+	+	+/-	–
Tbx18	+	–	–	–	–	–	–	–
Tbx20	+	+	+	+	+	+	+	+

Fig. 52.13 The early heart: distribution of the T-box transcription factors involved in patterning the heart.

disposition of the conducting system in a number of congenital anomalies. The concept accounts particularly well for the morphology and disposition of the atrioventricular node and bundle in hearts with straddling tricuspid valves, with double inlet left ventricles, and with congenitally corrected transposition.

SEPTATION OF THE EMBRYONIC CARDIAC COMPARTMENTS

The flow of blood on the right and left sides does not intermingle, even in the developing heart, because flow is laminar, and the pressures are similar at the right and left sides. It is only after birth that the pressures in the ‘left’ systemic circulation become higher than those in the ‘right’ pulmonary circulation; therefore, the different flows have to be separated physically in order to prevent admixture of oxygenated and deoxy-

genated blood. Prior to birth, the lungs are not functioning and blood flow to them is small. Even so, normal development of the heart requires that all cardiac compartments receive an adequate amount of blood. To this end, even before birth, the systemic blood of the right atrium is guided to the left atrium so that the left half of the heart and the brain is provided with a normal flow of blood. The right ventricle, in contrast, drains to the dorsal aorta via the ductus arteriosus (see Fig. 36.9B). The walls that separate the right and left sides of the heart are largely formed in the fourth and fifth weeks of development. Some time later, not only the components of the primary heart tube – specifically, the atrioventricular canal and the outflow tract – but also the atrial and ventricular chambers, must be separated physically into right and left halves. However, each compartment is separated in markedly different fashion.

Septation of the atrial chambers

As a prelude to atrial septation, the right horn of the sinus venosus is incorporated into the right part of the atrium, and the pulmonary vein arises as a midline structure within the dorsal mesocardium. Internal separation into right and left atria is mainly effected by growth of two septa, coupled with remodelling of the venous compartments. At the end of the fourth week of development, a crescentic fold, the primary atrial septum or septum primum, grows by active proliferation from the dorsocranial atrial roof into the direction of the atrioventricular canal, extending downwards just to the right of the entrance of the pulmonary vein (Fig. 52.14A,D). It is separated from the left sinuatrial valve by the interseptovalvular space. The leading edge of the septum primum is covered by a mesenchymal cap in continuity dorsally with extracardiac mesenchyme (mediastinal mesenchyme) derived from the dorsal mesocardium which forms the vestibular spine, or spina vestibule, also called the dorsal mesenchymal protrusion. The ventral horn of the septum primum reaches the ventral (cranial) atrioventricular cushion, while the dorsal horn reaches the dorsal (caudal) cushion. Thus the primary atrial foramen, the foramen or ostium primum, is entirely surrounded by mesenchyme derived from endocardium. Ventral and caudal to the advancing edge of the septum primum, the developing atrial chambers communicate through the foramen primum (see Fig. 52.14A,D).

As the foramen primum diminishes in size, the upper margin of the septum primum perforates by apoptosis and, thus, right to left shunting of blood continues through a secondary foramen, the foramen or ostium secundum. The latter is formed before the end of the fifth week of development. Initially, it lacks an upper rim, since its upper border is the atrial roof. The foramen primum is finally occluded in the median plane, as the edge of the septum primum merges with the fusing atrioventricular cushions (Fig. 52.14B,E). The foramen secundum then enlarges to permit free passage of blood from the right to the left atrium.

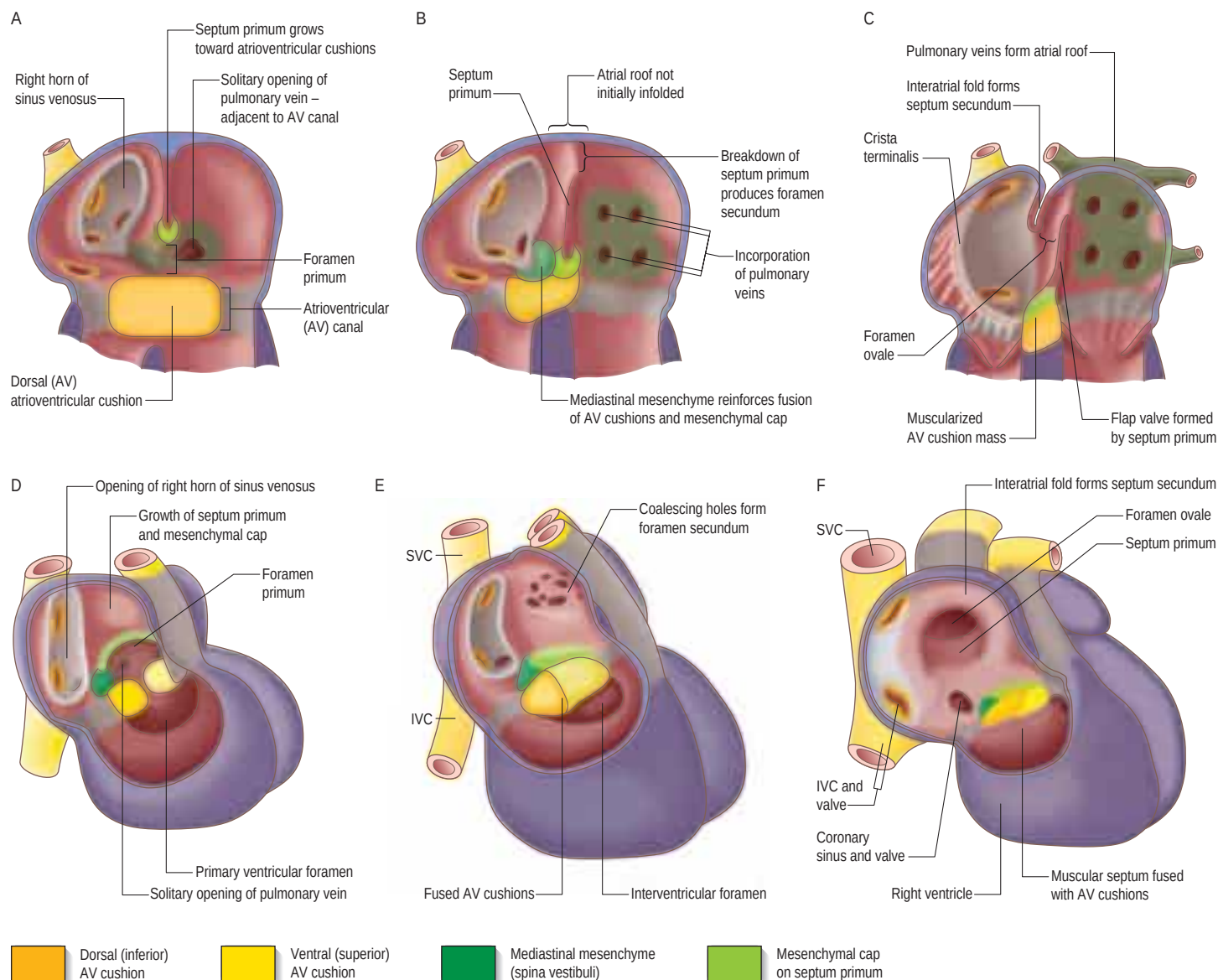


Fig. 52.14 Three representative stages in the development of the atrial septum. **A–C**, Coronal views from the front. **D–F**, Lateral views from the right side. **A** and **D**, The septum primum has not yet obliterated the original communication between the two atria (foramen primum) and the atrioventricular endocardial cushions have not yet fused. **B** and **E**, The atrioventricular endocardial cushions have fused with each other and with the septum primum, which has broken down in its dorsal part to form the foramen secundum. **C** and **F**, The septum secundum is forming on the right side of the foramen secundum from an interatrial fold. Abbreviations: AV, atrioventricular; IVC, inferior vena cava; SVC, superior vena cava.

The solitary opening of the pulmonary vein into the roof of the left atrium, which is originally adjacent to the atrioventricular junction, becomes incorporated into the atrial wall, ultimately forming four separate orifices at the corners of the atrial roof. As this development continues, the muscular atrial roof itself invaginates on the right side of the primary atrial septum (see Fig. 52.14B,E), the invaginated fold constituting the septum secundum. The free edge of the septum secundum overlaps the foramen secundum. When left atrial pressure exceeds right atrial pressure in postnatal life, the septum primum closes against the septum secundum as a flap-valve (Fig. 52.14C,F). The passage taken by blood as it passes from the right atrium – under the crescentic edge of the septum secundum, then obliquely towards and through the foramen secundum to the left atrium – persists throughout intrauterine life as the foramen ovale. At first, the foramen ovale is sited craniodorsally relative to the atrial septum, but with remoulding of the venous components, it achieves a cranioventral position.

Septation and appropriate positioning of the atrioventricular canal

Extracellular matrix accumulates between the endocardium and myocardium of the primary heart tube. However, it mostly disappears in the regions of ballooning of the chamber myocardium of the developing atria and ventricles. The matrix becomes filled with mesenchyme in the persisting regions of the primary heart tube. In conventional accounts

of the process, these regions are called the atrioventricular canal and the outflow tract, or conotruncus. The dorsal or inferior atrioventricular cushion continues into the floor of the atrium, which is made of primary myocardium. As the cushion extends dorsally, it meets the mediastinal mesenchyme in the region initially termed the spina vestibuli, where it continues as a mesenchymal cap running along the leading edge of the septum primum until it meets the ventral, or cranial, atrioventricular cushion (see Figs 52.14, 52.16). The dorsal cushion also has a significant ventricular extension in the inner curvature of the heart tube, which comes to lie on top of the developing muscular ventricular septum. The two atrioventricular cushions fuse in the sixth week of development, dividing the atrioventricular canal into its right and left components. The cushions are very large relative to the canal, leaving narrow right and left slits, which increase markedly in size during further development.

At the start of stage 10, the venous pole, the atrioventricular canal, the developing left ventricle and the outflow tract are all positioned symmetrically around the midline (Fig. 52.15A). As a result of the subsequent rapid growth of the cardiac tube, the atrioventricular canal moves in its entirety to the left (Fig. 52.15B) as the ventricular part of the heart tube loops to the right, thus placing the developing left ventricle on the left and the forming right ventricle on the right (see Fig. 52.9). This ventricular loop is conspicuous throughout the fourth and fifth weeks of development, and it is at this stage that a deep interventricular groove appears externally. Over this period, the atrial floor, including the developing systemic venous sinus (sinus venosus) and the

atrioventricular canal, moves to the right (Fig. 52.15C). The developing pulmonary vein remains anchored in the midline and, subsequent to these manoeuvres, the atrioventricular canal once more becomes positioned in the midline; this process facilitates the appropriate connections of the developing muscular septum with the atrioventricular and outflow cushions. The orifice of the sinus venosus is, by now, positioned to the right. From the outset, the muscular ventricular septum develops in line with the right side of the dorsal (inferior) atrioventricular cushion (Fig. 52.16), so that the separating right atrium always has direct access to the developing right ventricle. The process also ensures that space remains ventrally for incorporation of the subaortic component of the outflow tract into the left ventricle.

Septation of the ventricles

After the looping of the heart tube, it is easy to gain the impression that the atrioventricular canal communicates exclusively with the developing left ventricle, while the outflow tract is supported exclusively by the developing right ventricle. In the definitive heart, blood entering the

right ventricle passes in through the right atrial vestibule and exits through the muscular subpulmonary infundibulum, while blood entering the left ventricle passes in through the left atrial vestibule, and exits through the aortic root. The changes needed to produce the definitive connections between the cardiac components are complex. Three distinct structures contribute to the formation of the postnatal ventricular septum: the muscular ventricular septum, the proximal parts of the outflow cushions and the atrioventricular endocardial cushions. The latter two components separate those parts of the primary heart tube that are committed to the right and left ventricles, as opposed to the ballooned apical ventricular components. Inappropriate formation and connection of the cushions with the muscular ventricular septum underscores deficiencies of the definitive ventricular septum; such lesions account for about two-thirds of all cardiac septal defects.

Separation between the right and left ventricles is initially heralded by the appearance of a caudal crescentic ridge within the ventricular loop. The trabeculated parts of the ventricles contain less extracellular matrix than the walls of the primary heart tube, and it is these parts of the chambers that expand on each side of the ridge that remains between them. The crest of the developing septum, therefore, is its oldest portion. The more apical parts are added concomitant with enlargement of the chambers, as if they were expanding like balloons. The impression can be gained that the dorsal and ventral horns of the ventricular septum grow along the ventricular walls, meeting and fusing with the right extremities of the dorsal and ventral cushions of the atrioventricular canal. In reality, the crest of the septum marks the position of the original primary heart tube and becomes the atrioventricular bundle. The septum has a free, sickle-shaped margin that, together with the fused caudal surface of the endocardial cushions, bounds an ovoid foramen. Previously, this foramen has erroneously been termed the interventricular, or bulboventricular foramen; it is no more than a locus within the cavity of the primary heart tube. The apical trabecular components of the ventricles balloon from the ventral side of the primary heart tube. At the dorsal side, or the inner curvature, there is no chamber myocardium, only the smooth walls of the primary heart tube. Therefore, from the outset of the process, the forming apical parts of the ventricles are separated by a muscular septum. The foramen marked caudally by the crest of the ventricular septum provides the initial inlet to the developing right ventricle, and the outlet for the developing left ventricle. Completion of ventricular septation requires division of this primary foramen, rather than its closure. It also demands appropriate connection between the muscular septum and the divided atrioventricular canal, including its cushions, and with the divided outflow tract, along with its cushions (see Fig. 52.16).

Use of inappropriate terminology means that the entire region upstream relative to the foramen has previously been called the primitive ventricle, the entire downstream region the bulbus, and the junction between them the bulboventricular junction. In the account presented here, the area is termed the primary junction, initially marked by a distinct notch on the outside of the heart, and inside by the corresponding ridge of the developing ventricular septum. The ridge is positioned between the atrioventricular orifice, which is initially a common structure, and the caudal part of the forming right ventricle.

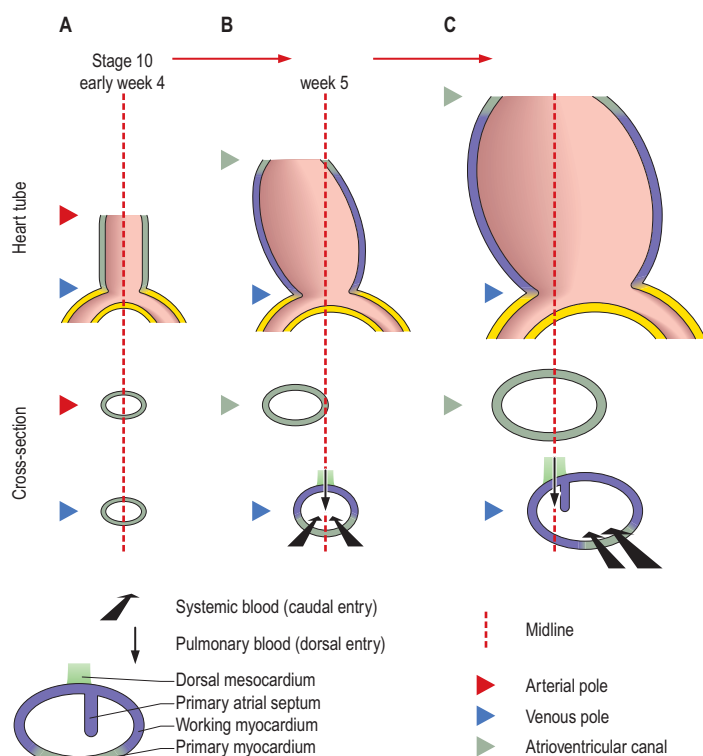


Fig. 52.15 Movements of the atrioventricular canal relative to the midline during development.

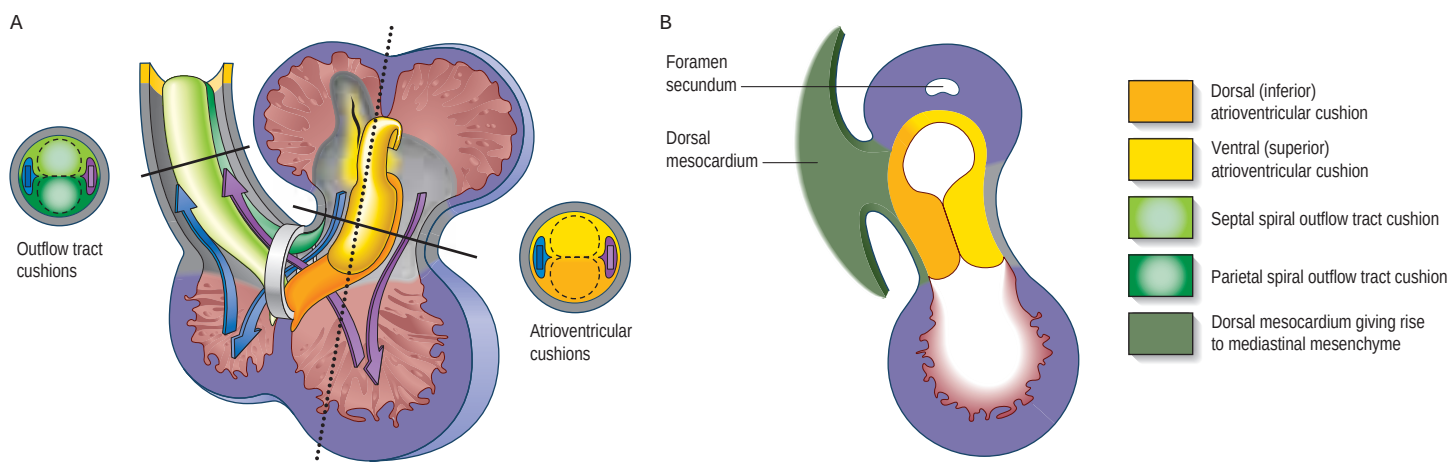


Fig. 52.16 Septation of the ventricles and outflow tract. **A**, A frontal view of the heart at stage 13, with the outflow tract reflected to the right. Lines show the level of transverse sections of the atrioventricular cushions and outflow tract. Note that blood from both atria passes through the embryonic interventricular foramen at this stage. **B**, A sagittal section of heart along the dotted line in **A**, showing the atrioventricular junctional region, seen from the right side. The atrioventricular cushions extend on to the leading edge of the septum primum (although the mesenchymal cap on the atrial septum – see Figure 52.14 – probably has a different developmental origin).

Appropriate connection with the separating atrioventricular and arterial junctions is essential for the development of a four-chambered heart. The serendipitous discovery of the expression of a 'neural' antigen in the primary junctional myocardium, surrounding the so-called primary interventricular foramen, has clarified the complex morphogenic steps involved in the remodelling of this region at the inner curvature and the final position of the original interventricular ring in the fully developed heart (see Fig. 52.12).

Septation and appropriate positioning of the outflow tract

The length of the myocardial portion of the outflow tract decreases markedly between the fourth and eighth weeks of development. In part, the myocardium becomes incorporated into the ventricles; in part, it disappears by apoptosis. In the definitive heart, there is no distinct myocardial outflow tract present within the left ventricle because the myocardium that initially separated the developing mitral and aortic valves disappears, leaving the fibrous aortic-mitral continuity distinctive for the postnatal heart. In the right ventricle, the proximal myocardial outflow tract persists as the smooth-walled muscular sub-pulmonary infundibulum. Formation of the part adjacent to the aortic root requires muscularization of the most proximal parts of the outflow cushions to form the greater part of the supraventricular crest, or crista supraventricularis.

Key to separation of the outflow tract is the appearance of the two outflow ridges, or cushions, which spiral one round another as they run from the distal end of the right ventricle, where they are positioned septally and parietally, to the aortic sac (see Fig. 52.16; Fig. 52.17). Within the aortic sac, a transverse wedge of tissue, termed the aorto-pulmonary septum, separates the origins of the arteries traversing the fourth and sixth pharyngeal arteries. The septum is formed from dorsal mediastinal mesenchyme. Myocardial precursor cells and non-myocardial cells are added to the outflow tract, the latter forming the intrapericardial components of the arterial trunks. Neural crest cells migrate from the pharyngeal mediastinal myocardium into the outflow cushions. They do not appear to populate the septum itself, but form the larger parts of the walls of the intrapericardial portions of the truncus arteriosus. The precise relationship between the outflow cushions and the newly formed aortic arch arteries, which never possess a septum between them, has still to be assessed.

The spiral outflow ridges start to fuse from the sixth week onwards, initially in the distal part of the outflow tract, and fusions continue gradually into the direction of the ventricular septum (see Fig. 52.17;

Fig. 52.18A). The most proximal parts of the cushions remain unfused when the aorta and pulmonary trunk have gained their separate existence within the pericardial cavity. At this time, the proximal part of the outflow tract, upstream to the intrapericardial arterial trunks, remains encased in a myocardial sleeve. The arterial valves and sinuses then develop within the more distal part of this myocardial outflow tract; the outer myocardial wall does not disappear completely until well after the completion of septation. The most proximal parts of the cushions are the last ones to fuse, and they then muscularize and join the muscular ventricular septum, thereby joining the aorta into the left ventricle. The dorsal atrioventricular cushion maintains its mesenchymal character, becoming the membranous part of the interventricular septum in the formed heart. Formation of its interventricular and atrioventricular components does not become evident until the septal leaflet of the tricuspid valve delaminates from the surface of the ventricular septum.

NON-MYOCARDIAL COMPONENTS OF THE HEART

The non-myocardial components of the heart are the epicardium, including the coronary vasculature, and the interstitial fibroblasts and the valvular apparatus, which are formed, in their larger part, from endocardially derived mesenchyme. From the outset, the atrioventricular valves, mitral and tricuspid, are formed at the site of the initial atrioventricular canal, whereas the aortic and pulmonary valves are initially developed within the myocardial outflow tract, and only later achieve the semilunar attachments of the leaflets, which cross the anatomical ventriculo-arterial junctions. All of the leaflets form initially as internal endocardial projections that enclose a myocardial basement membrane, matrix and mesenchymal cells. The precise mechanisms involved in the formation of these areas have still to be determined.

Development of the epicardium and the coronary vasculature

The epicardium, coronary vascular bed and interstitial fibroblasts develop from a mesothelially covered protrusion of mesenchymal cells, the pro-epicardium, which arises from the pericardium in the region of the sinus venosus during week 5. The base of the pro-epicardium encompasses bi-potential pericardial cells, which are recruited either to the cardiac lineage to form the venous pole of the heart, or else to the epicardial lineage. The entire structure expresses Tbx18. Bone morphogenetic proteins and fibroblast growth factors influence which lineage is followed; the precise mechanism of specification is not yet clear. Cells

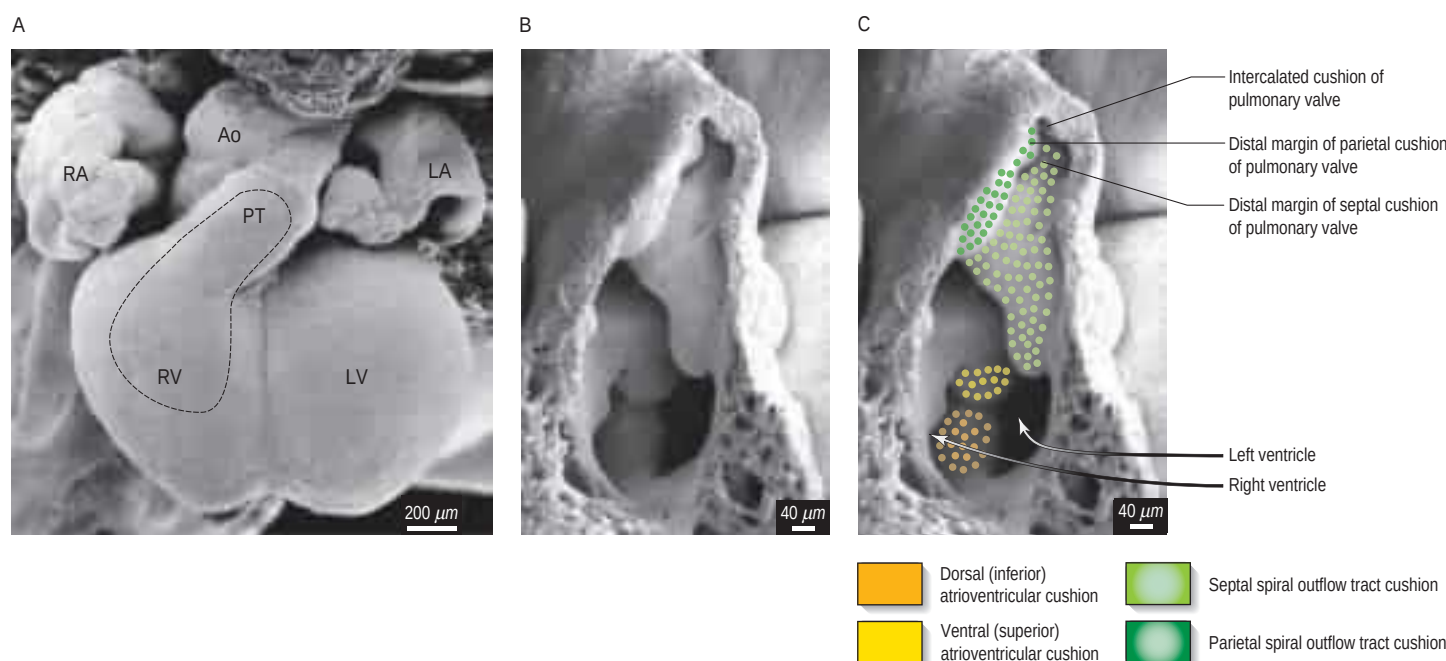


Fig. 52.17 Septation of the outflow tract. **A**, A scanning electron micrograph (SEM) of a human heart at stage 18. The pulmonary trunk is separate from the aorta. **B**, An SEM of the human heart at stage 17, showing the luminal aspect of the right ventricle and the outflow tract. The anterior and parietal walls of the outflow tract have been removed along the dotted line shown in **A**. **C**, An SEM with added colour to show the structures in **B**. The outflow cushions are unfused. The distal margins of both outflow tract cushions are contributing to the developing pulmonary valve: the third leaflet and sinus of this valve are derived from the dorsal aspect of the outflow tract cushions. The components of the aortic valve develop from the dorsal aspect of the outflow tract cushions. Abbreviations: Ao, aorta; LA, left atrium; LV, left ventricle; PT, pulmonary trunk; RA, right atrium; RV, right ventricle. (Courtesy of Prof G Steding, Göttingen, with permission from Oostra RJ, Steding G, Lamers WH, Moorman AFM. Steding's and Virágh's Scanning Electron Microscopy Atlas of the Developing Human Heart. New York: Springer-Verlag, 2006.)

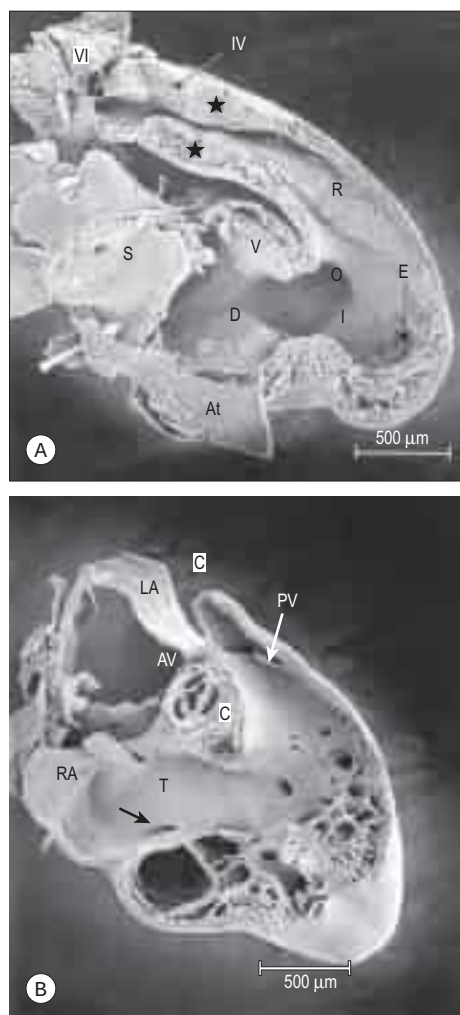


Fig. 52.18 **A**, A scanning electron micrograph (SEM) of a human heart at 32 days (stage 15). The right lateral wall of the right atrium, right ventricle and the outflow tract have been removed. The septum primum (S) and the developing valve of the sinus venosus (white arrow) are at a distance from the unfused ventral (V) and dorsal (D) atrioventricular cushions. The interventricular foramen (primum) is open (O). Its lower margin is made up by the developing interventricular septum (I). Note the left anterolateral ridge (R) and, at the distal margins of the truncus arteriosus, the valve swellings (*) of the putative aorta and pulmonary trunk. Other abbreviations: At, atrium; E, embryonic trabeculae; IV, origin of fourth aortic arch; VI, origin of sixth aortic arch. **B**, The septal aspect of the right ventricle of a human heart at 8 weeks. Delamination of the septal cusp of the tricuspid valve (T) is taking place. A cleft (black arrow) has appeared, marking the posterior margin of the septal cusp. There are no clefts delineating the anterior margins in the region adjacent to the supraventricular crest (C), and the endocardium is continuous. No tension apparatus (chordae tendinae and papillary muscles) is present. Other abbreviations: AV, aortic valve; LA, left atrium; PV, pulmonary valve; RA, right atrium.

detach from the pro-epicardium and spread over the myocardial tube, caudally to cranially and dorsally to ventrally. Thus, in successive stages, the pro-epicardial cells approach the myocardial surface of the developing atria, atrioventricular canal, ventricles and the proximal part of the outflow tract. The distal part of the outflow tract, as it becomes arterialized with regression of the initial myocardial walls, is covered by an outgrowth of the pericardial mesothelium at the site of the initial pericardial reflections. After the formation of the epicardium, a space is formed between the developing epicardium and the outer myocardial layer, which is particularly pronounced at the atrioventricular and outflow regions of the heart tube. This is the subepicardial space, or subepicardium; it is subsequently populated by mesenchymal cells that are largely derived from the epicardium and which play an important role in the formation of the coronary vasculature. The precursors of coronary fibroblasts, smooth muscle cells and endothelium are all derived from these pluripotent cells, which possibly also play roles in the development of non-vascular tissues, including the mesenchyme of the atrioventricular cushions and some myocardial elements. The remnants of the pro-epicardium can be found in the area of the developing systemic venous tributaries up to stage 15.

The coronary arteries were previously assumed to form by means of angiogenesis, or sprouting from the aortic root. However, recent studies have demonstrated that the vessels develop prior to the formation of the arterial orifices in the aortic sinuses; it is now thought that the vessels arise by vasculogenesis, and that mesenchymal cells in the sub-epicardial space give rise to primitive vessels that invade the myocardium and spread over the myocardial surface. Sprouts of this plexus approach the base of the outflow tract and connect to the sinuses of the aortic root. Ultimately, only solitary right and left coronary arteries remain, although, in up to one-third of human hearts, small accessory branches of the coronary arteries take origin directly from the coronary sinuses. The vessels that remain acquire a medial wall and an adventitial covering, whereas those that do not form a medial wall are destined to regress. The coronary veins develop by sprouting from the coronary sinus, draining directly to the atrial chambers. The mature coronary vascular pattern is established by remodelling of the capillary plexuses and reduction in the number of arteriovenous anastomoses.

Atrioventricular valves

Initially, the atrial myocardium is continuous with the ventricular myocardium through the myocardium of the atrioventricular canal. The atrioventricular canal itself is characterized by the tissues of the atrioventricular groove on the epicardial side, and those of the endocardial cushions on the endocardial side. As the atrioventricular valves develop, these tissues fuse at the ventricular margin of the canal, and so disrupt myocardial continuity. Although molecular lineage studies in the mouse have demonstrated that the atrioventricular cushions are initially entirely populated by endocardium-derived mesenchymal cells, they receive a significant epicardial contribution during further development, the fibroblasts making up the valvular leaflets, and their tendinous cords in the formed heart are therefore derived from both the endocardium and the epicardium.

The atrioventricular valves develop as shelf-like projections from the margins of the atrioventricular orifices. They are directed as almost complete conical sheets towards the ventricles, their advancing edges continuing, initially as trabecular ridges, deep into the ventricular cavity. With continued differential growth and excavation on their ventricular aspects, each sheet develops major indentations forming the mitral and tricuspid orifices and defining their principal leaflets. Minor marginal indentations subdivide some leaflets into scallops. Each leaflet develops functionally significant regional variations in surface texture and its core condenses as a collagenous fibrous lamina, supported either by the fibro-fatty tissue of the atrioventricular groove, or by a fibrous anulus. The anterior leaflet of the tricuspid valve and the aortic and mural leaflets of the mitral valve all appear at about the time of fusion of the atrioventricular and outflow cushions. Delamination of the septal leaflet of the tricuspid valve occurs after the completion of ventricular septation, producing the interventricular and atrioventricular components of the membranous septum, and continuing until at least week 11 (Fig. 52.18B).

Aortic and pulmonary valves

The aortic and pulmonary valves form from endocardial cushions within the distal end of the myocardial outflow tract and become populated by neural crest cells, in addition to endocardial- and pharyngeal-derived mesenchyme. The fusing outflow cushions divide, providing two rudiments each for the developing aortic and pulmonary valves. The remaining leaflet for each valve is then produced from the newly appearing intercalated cushions (see Fig. 52.17). Each cushion-derived intrusion grows, and is also excavated on its mural aspect to form a semilunar valvular leaflet. Similar events affect the adjacent wall. In this way, the pouches between the valves and the walls of the cushions gradually enlarge, the mural aspect of each cushion forming the valvular sinuses. The core of each leaflet forms a collagenous fibrous lamina, delicate and thin in each crescentic lunule, thick and compact in the central nodule, with marginal radiate and basal bands. The dynamic nature of cardiac development makes it difficult to describe the positions of the leaflets precisely in space; it is, therefore, best to describe the right, left and non-coronary sinuses and leaflets of the aortic valve, and the adjacent and non-adjacent leaflets of the pulmonary valve.

FETAL CIRCULATION

As the cardiovascular system develops, the embryonic circulation (Ch. 13) is modified into a fetal circulation, which maintains a placental

circulation and is also able to establish the pulmonary circulation at birth. Extensive remodelling of the early aortic arch arteries and (initially bilateral) dorsal aortae occurs (Ch. 36). The dorsal aortae persist on the cranial side of the third aortic arches as continuations of the internal carotid arteries (see Fig. 36.9). The dorsal aorta between the third and fourth aortic arches, the carotid duct, diminishes and finally disappears. From the fourth arch to the origin of the seventh intersegmental artery, the right dorsal aorta becomes part of the right subclavian artery, and caudal to the seventh intersegmental artery, the right dorsal aorta disappears as far as the locus of fusion of the thoracic aortae. After disappearance of the left carotid duct, the remainder persists to form the descending part of the arch of the aorta. The fused right and left embryonic dorsal aortae persist as the definitive descending thoracic and abdominal aorta. A constriction, the aortic isthmus, is sometimes present in the aorta between the final site of origin of the left subclavian artery and the orifice of the arterial duct.

Concomitant changes also occur in the early venous system. The precardinal veins enlarge as the head and brain develop. They are further augmented by the subclavian veins from the upper limb buds, which become the chief tributaries of the common cardinal veins, gradually assuming an almost vertical position as the heart descends into the thorax (see Fig. 13.4). That part of the original precardinal vein rostral to the subclavian vein is now the internal jugular vein, and their confluence is the brachiocephalic vein of each side. The right and left common cardinal veins are originally of the same diameter; by the development of a large oblique transverse connection, the left brachiocephalic vein carries blood across from the left to the right. The part of the original right precardinal vein between the junction of the two brachiocephalic veins and the azygos veins forms the upper part of the superior vena cava. The caudal part of this vessel, below the entrance of the azygos vein, is formed by the right common cardinal vein. The left precardinal and left common cardinal veins caudal to the transverse branching of the left brachiocephalic vein largely atrophy; the precardinal constitutes the terminal part of the left superior intercostal vein, while the common cardinal is represented by the ligament of the left superior vena cava and the oblique vein of the left atrium. The remainder of the left superior intercostal vein is developed from the cranial end of the postcardinal vein and drains the second, third and, on occasion, the fourth intercostal veins. The oblique vein passes downwards across the back of the left atrium to open into the coronary sinus, which represents the persistent left horn of the systemic venous sinus.

The fetal circulation contains a number of relatively large vessels that permit the majority of the blood flow to bypass the liver and lungs. The placenta serves as the organ for fetal nutrition and excretion, receiving deoxygenated fetal blood and returning it oxygenated and detoxified. Fetal blood reaches the placenta via two umbilical arteries and, in early fetal life, returns by two umbilical veins (Fig. 52.19). The right umbilical vein later disappears, whereas the persisting left umbilical vein enters the abdomen at the umbilicus, traverses the edge of the falciform ligament to reach the hepatic surface, and then joins the left branch of the portal vein at the hepatic portal. Opposite the junction, a large vessel, the ductus venosus, arises and ascends posterior to the liver to join the left hepatic vein near its termination in the inferior vena cava. (For a detailed developmental account of the circumhepatic veins, see Figs 60.8–60.11.) The portal vein is small in the fetus compared to the size of the umbilical vein. Parts of the left branch of the umbilical vein, proximal and distal to their junctions, function as branches of the portal vein, carrying oxygenated blood to the right and left parts of the liver. Blood in the left umbilical vein, therefore, reaches the inferior vena cava by three routes: some enters the liver directly and reaches the vena cava via the hepatic veins; a considerable quantity circulates through the liver with portal venous blood before also entering by the hepatic veins; and the remainder is bypassed into the inferior vena cava by the ductus venosus.

The refreshed placental blood passes almost directly to the aorta for distribution to the head and upper limbs. Blood from the ductus venosus and hepatic veins mixes in the inferior vena cava with blood from the lower limbs and abdominal wall, and enters the right atrium. Because right atrial pressure is much greater than left atrial pressure, it forces the flap-like valve of the septum primum to the left, which permits passage of blood from the right to the left atrium. The valve of the inferior vena cava is so placed as to direct 75% of the richly oxygenated blood from the umbilical vein to the foramen ovale and left atrium, where it mingles with the limited venous return from the pulmonary veins. From the left atrium, blood enters the left ventricle and from there flows to the aorta; from here, it is probably distributed almost entirely to the heart, head and upper limbs, which means that little reaches the descending aorta.

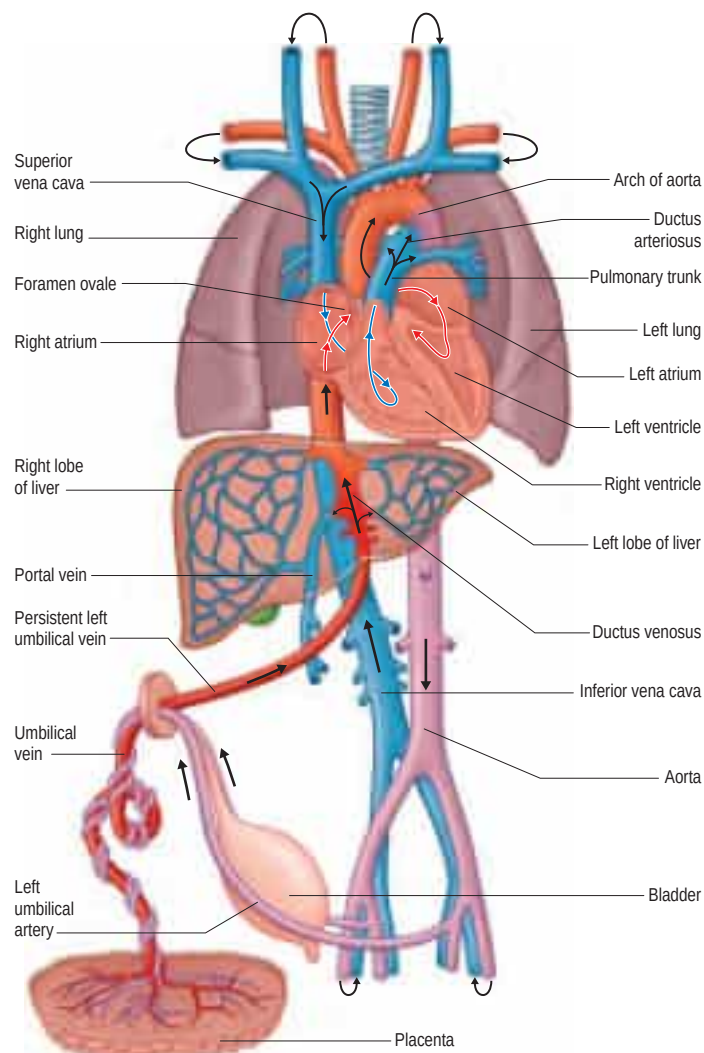


Fig. 52.19 A plan of the fetal circulation. The arrows indicate the direction of blood flow. The placenta is drawn to a greatly reduced scale.

Blood from the head and upper limbs returns to the right atrium via the superior vena cava, flows through the right atrioventricular orifice into the right ventricle (together with the small amount of blood that is returned via the inferior vena cava), and then passes into the pulmonary trunk. However, because the fetal lungs are largely inactive, only a little of the blood from the right ventricle flows through the right and left pulmonary arteries and returns to the left atrium via the pulmonary veins. The greater part of the outflow through the pulmonary trunk is carried by the ductus arteriosus directly to the aorta, where it mixes with the small quantity of blood that passes from the left ventricle into this part of the aorta. The mixture descends in the aorta and most is returned via the umbilical arteries to the placenta; some is distributed to the lower limbs and the organs of the abdomen and pelvis.

CHANGES IN THE FETAL CIRCULATION AND OCCLUSION OF FETAL VESSELS AFTER BIRTH

At birth, as pulmonary respiration begins, increased amounts of blood from the pulmonary trunk flow through the pulmonary arteries to the lungs and return by the pulmonary veins to the left atrium; pressure therefore increases within the left atrium. A decrease in pressure also occurs in the inferior vena cava as a result of the reduction of venous return concomitant with occlusion of the umbilical vein and ductus venosus. Atrial pressures become equal and the valvular foramen ovale is closed by apposition, and subsequent fusion, of the septum primum to the rims of the foramen. Contraction of the atrial septal muscle, synchronized with that in the superior vena cava, may assist this closure, which occurs after functional closure of the ductus arteriosus. Although the foramen ovale closes functionally after pulmonary respiration is established, it does not become structurally closed until some time later. It is obliterated in fewer than 3% of infants 2 weeks after birth, and in 87% by 4 months after birth. Fusion is sometimes incomplete

and a potential atrial communication (atrial septal defect) persists throughout life. Almost always, this has no functional effect because the inequality of atrial pressures and the valve-like arrangement of the opening do not favour passage of blood.

Soon after birth, a number of fetal vessels occlude, although the majority remain patent. This differential constriction suggests that the walls of a population of fetal vessels are different to those of the remaining vessels. Bradykinin, one of the kinin polypeptide hormones that induce contraction or relaxation of smooth muscle, forms in the blood of the umbilical cord when the temperature of the cord decreases at or shortly after birth. It is also formed and released by granular leukocytes in the lungs of the neonate after exposure to adequate oxygen. Bradykinin is a potent constrictor of the umbilical arteries and veins, and of

the ductus arteriosus, and is also a potent inhibitor of contraction of the pulmonary vessels.

Ductus arteriosus

The ductus arteriosus shunts blood from the pulmonary trunk to the arch of the aorta, bypassing the fetal lungs (see Fig. 52.19; Fig. 52.20). It arises as a direct continuation of the pulmonary trunk at the point where it divides into right and left pulmonary arteries. It is 8–12 mm long, and joins the aorta at an angle of 30–35° on the left side, antero-laterally, below the origin of the left subclavian artery. The opening of the ductus arteriosus into the aorta is greatly elongated. Its diameter at

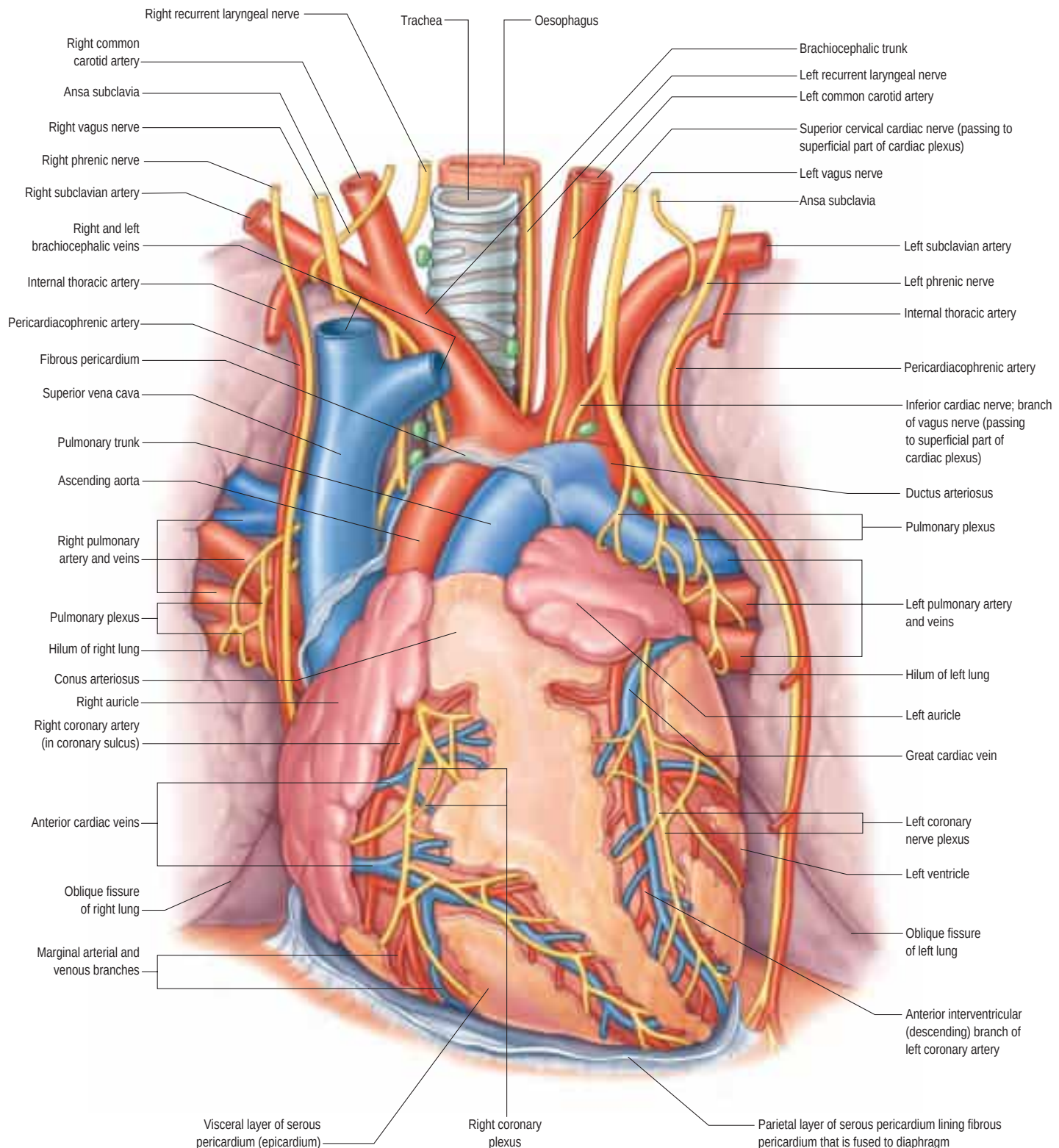


Fig. 52.20 An anterior view of the heart and great vessels in a full-term neonate. The lungs have been displaced to expose the heart, and the epicardium has been dissected off the heart and the roots of the great vessels. Note that, after birth, blood flow reverses through the ductus arteriosus prior to its closure. (After Crelin ES 1969 *Anatomy of the Newborn*. Philadelphia: Lea and Febiger.)

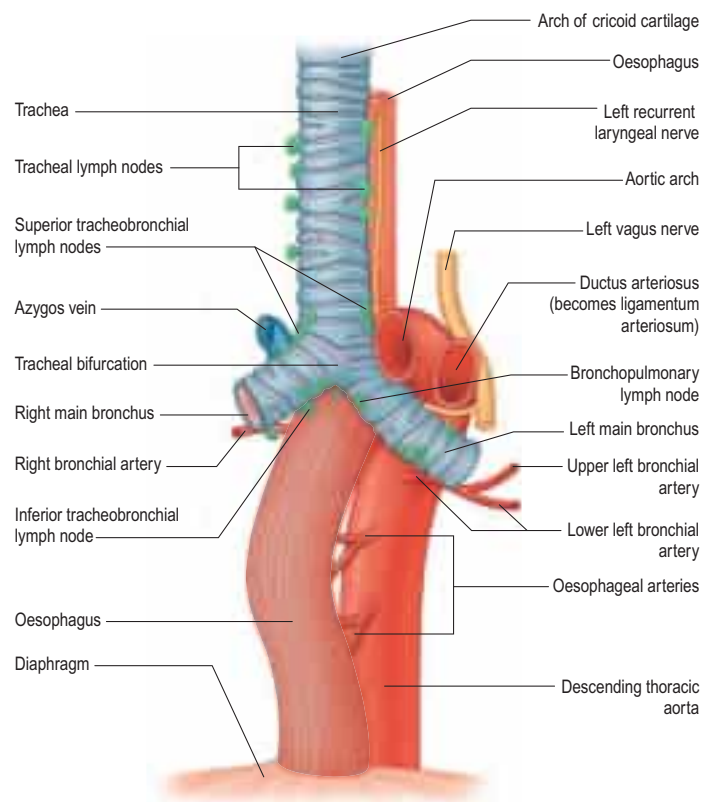


Fig. 52.21 An anterior view with the heart removed to show the relationship between the left primary bronchus, the aortic arch and the ductus arteriosus in a full-term fetus. (After Crelin ES 1969 *Anatomy of the Newborn*. Philadelphia: Lea and Febiger.)

its origin from the pulmonary trunk, when distended with blood, is 4–5 mm, which is nearly equal to the diameter of the adjacent ascending aorta (5–6 mm). Both arteries taper to a smaller diameter as they pass inferiorly and the aorta remains slightly larger (4 mm) (Fig. 52.21). In the neonate, the ductus arteriosus is closely related to the left primary bronchus inferiorly and the thymus gland anteriorly.

The ductus arteriosus differs from the other great vessels that arise from the heart because its tunica media contains smooth muscle rather than layers of elastin. It has been suggested that the relationship between the recurrent laryngeal branch of the vagus nerve and the developing ductus arteriosus is responsible for this histological difference on the grounds that the left recurrent laryngeal nerve (which, in a stage 16 embryo, is very large in relation to the aortic arch system) supports the ductus arteriosus as it envelops it, thereby permitting the ductus to develop as a muscular, and not an elastic, artery.

It is essential that the ductus arteriosus remains patent during intra-uterine life. Prostaglandins appear to have a role in maintaining this patency. Fetal and neonatal ductal tissue can produce prostaglandins E_2 , I_2 and F_{2a} , which inhibit the ability of the ductus to contract in response to oxygen.

Closure of the ductus arteriosus

The ductus arteriosus starts to close immediately after birth, although blood probably continues to flow intermittently through it for a week or so. The flow is reversed relative to that occurring in the fetal circulation, reflecting the increased systemic vascular resistance that follows exclusion of the placental circulation, and the decreased pulmonary resistance that follows expansion of the lungs. Animal studies have shown that reversal of flow occurs rapidly after ventilation begins. Blood flow from the left ventricle travelling via the aorta and ductus arteriosus contributes almost 50% of total pulmonary blood flow. Initial constriction at birth has been attributed to increased oxygen tension. A neural factor may also be involved; the muscular wall has afferent and efferent nerve endings and responds to adrenaline (epinephrine) and noradrenaline (norepinephrine) within 10 minutes of ventilation onset; this reduces to about 25% by 2 hours. It is suggested that the turbulence and increased shear-stress of this blood flow releases vasoconstrictive factors from the endothelium (Crossley et al 2009).

The first stage of ductal closure is completed within 10–15 hours and the second stage takes 2–3 weeks. The first stage consists of contraction of the smooth muscle cells and development of subendothelial oedema.

Destruction of the endothelium and proliferation of the intima subsequently occur, and lead to permanent closure. Diverse factors that may promote ductal closure have been identified, and include increased oxygen tension; increased plasma catecholamine concentrations; suppression of prostaglandin I_2 production; switching off prostaglandin E receptors; a synergistic role of prostaglandin F_{2a} and oxygen concentrations; and a decrease in plasma adenosine concentration. After birth, these interrelated events result in the closure of the ductus arteriosus. It has been proposed that the high oxygen tension of the reversed blood flow through the ductus initiates the synthesis of a hydroperoxy fatty acid that suppresses prostacyclin production, thus exposing the ductus to the contractile effects of prostaglandin endoperoxide. After closure, the duct becomes the ligamentum arteriosum, which connects the left pulmonary artery (near its origin) with the aortic arch.

FETAL AND NEONATAL HEART

At full term, the heart is situated midway between the crown of the head and the lower level of the buttocks (see Fig. 14.6). The anterior surface is formed mainly by the right atrium and right ventricle, as it is in the adult. This surface is usually covered by the thymus, which may extend over the base of the right ventricle. The heart is relatively large at birth, weighing around 20 g. The cardiac output is around 550 ml per minute and the blood pressure is 80/46 mmHg. The heart rate during fetal life, as term approaches, is around 150 beats per minute (bpm). It increases at birth to around 180 bpm, decreases over the first 10 minutes after birth to 170 bpm, and reaches 120–140 bpm from 15 minutes to 1 hour after birth. Any signs of fetal distress will increase this general basic level. The heart rate decreases further with increasing age: it is normally 113–127 bpm from 6 months to 1 year, and settles to around 100 bpm by the end of the first year.

Considered relative to the thoracic landmarks, the foramen ovale lies at the level of the third intercostal space, with its long axis in the median plane. It is 4–6 mm in vertical length and 3–4 mm wide (see Fig. 14.6). It is almost exactly in the coronal plane of the body, so that blood passes from the anterior, or ventral, right atrium posteriorly and upwards to reach the upper and posterior part of the left atrium. After birth, the intra-atrial pressures are equalized, and the free edge of the flap valve formed by the primary atrial septum is kept in contact with the left side of the rims of the fossa, promoting subsequent anatomical fusion, despite the fact that the foramen remains probe-patent in up to one-third of all individuals. The initially free crescentic margin of the infolded superior interatrial fold forms the border of the fossa after fusion; the flap valve formed by the primary septum accounts for its floor in the adult heart.

Functionally, the ventricular septum is often considered to be part of the left ventricle; while this is certainly true in terms of the orientation of the myocardial aggregates that make up the ventricular walls, it is not correct from a cellular lineage perspective. The ratio of cardiac weight is usually expressed as the weight of the right ventricle relative to that of the left ventricle and the septum. Calculated in this fashion, the left ventricle at birth weighs about 25% more than the right ventricle. However, the right ventricle has been working against the systemic pressure in the fetus, the pulmonary circulation not yet being active, and there is a preponderance of right ventricular function in the first 2 or 3 months after birth. With the establishment of the pulmonary circulation, the work of the right side of the heart decreases, and the left side of the heart, particularly the ventricle, grows rapidly to meet the demands of the active neonate. By the end of the second year, the left side weighs twice as much as the right, a ratio that continues to middle age. At birth, the average thicknesses of the lateral walls of the ventricles are approximately equal (5 mm). By the end of the third postnatal month, the left ventricle has already become thicker than the right; it becomes twice as thick by the second year, and three times as thick by puberty.

CONGENITAL HEART DEFECTS

Defects of cardiac development may not come to light until after birth, although many, if not most, can now be observed on fetal ultrasonic examination. They may affect any part of the developing heart, and include more complex forms with inappropriate connections of cardiac components. They include anomalies such as double inlet ventricle; absence of one atrioventricular connection, commonly described as tricuspid or mitral atresia; and discordant atrioventricular and ventriculo-arterial connections, also known as congenitally corrected transposition.

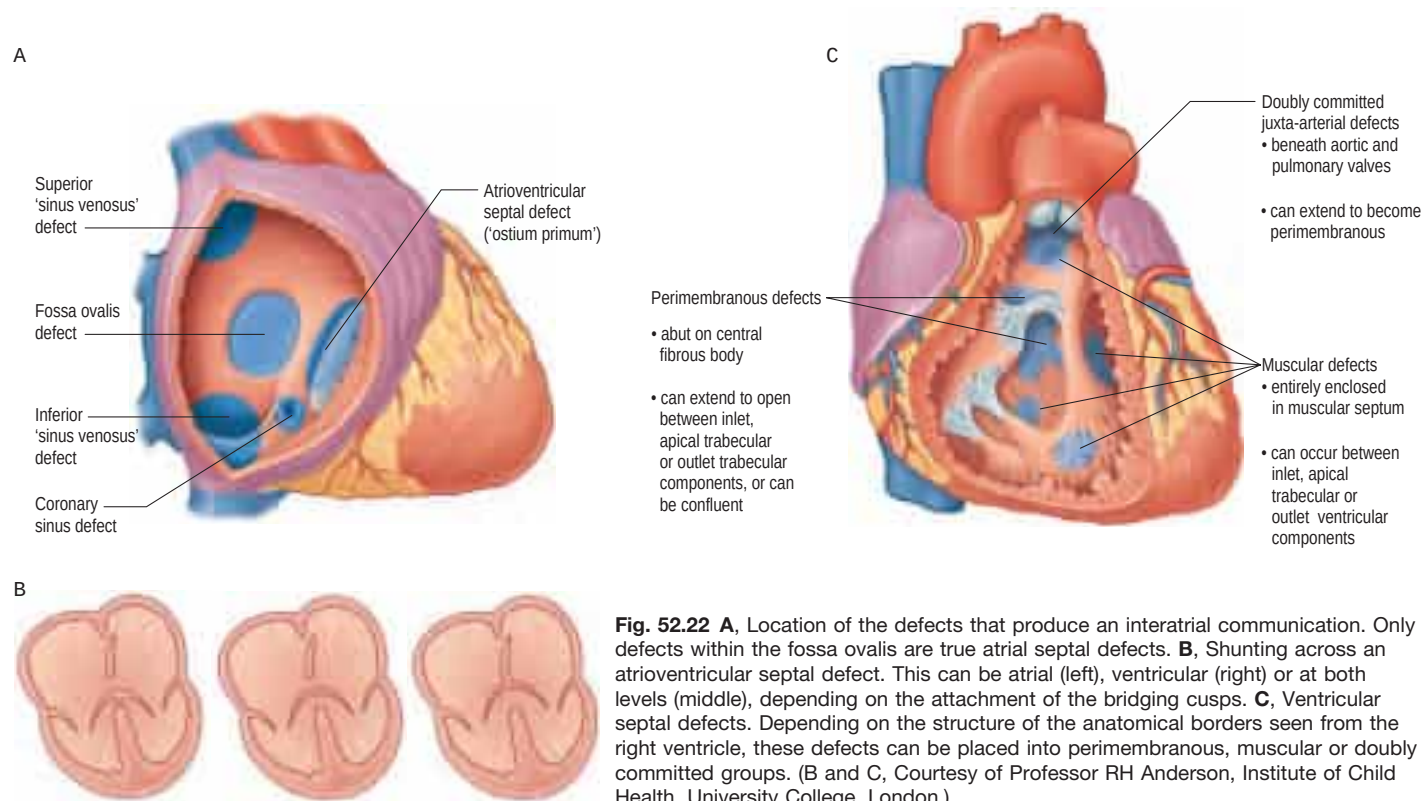


Fig. 52.22 **A**, Location of the defects that produce an interatrial communication. Only defects within the fossa ovalis are true atrial septal defects. **B**, Shunting across an atrioventricular septal defect. This can be atrial (left), ventricular (right) or at both levels (middle), depending on the attachment of the bridging cusps. **C**, Ventricular septal defects. Depending on the structure of the anatomical borders seen from the right ventricle, these defects can be placed into perimembranous, muscular or doubly committed groups. (B and C, Courtesy of Professor RH Anderson, Institute of Child Health, University College, London.)

Congenital cardiac anomalies are often multiple, and probably occur more frequently in siblings and in children of consanguineous marriages; there is a low correlation among monozygotic twins. Ventricular septal defects are the most common lesions, making up around one-fifth of all cases, followed by persistent patency of the ductus arteriosus, coarctation, pulmonary stenosis, Fallot's tetralogy, transposition, aortic stenosis, and hypoplasia of the left heart.

Atrial septal defects

A persistent communication between the atrial chambers within the fossa ovalis is common, and results from the failure of the flap valve of the primary atrial septum to fuse with the infolded muscular rims of the fossa. When the flap valve is still able to overlap the rims, the communication is of no functional significance as long as left atrial pressure is greater than right, which is usually the case. However, when the flap valve is smaller than the fossa ovalis, or when it is perforate, there is a true atrial septal defect (Fig. 52.22A).

In normal development, the free leading edge of the septum primum merges with the atrioventricular endocardial cushions, permitting subsequent formation of the atrioventricular septum. When this does not happen, there is a common atrioventricular junction and an atrioventricular septal defect. This can occur when the leaflets of the atrioventricular valves are fused to the crest of the ventricular septum (see Fig. 52.22A), producing an interatrial communication at the expected site of the atrioventricular septum. This is the so-called ostium primum defect; in reality, it is an atrioventricular septal defect permitting exclusive atrial shunting. Other interatrial communications can be formed in the mouths of the vena cavae, most frequently the superior vena cava, and are usually associated with drainage of the right pulmonary veins into the cavo-atrial junction. Known as sinus venosus defects (see Fig. 52.22A), their essential feature is a bi-atrial connection of the involved caval vein. An interatrial communication can also occur through the mouth of the coronary sinus when there is a deficiency or absence of the walls that usually separate the sinus from the left atrium.

Atrioventricular septal defects result from failure of fusion of the endocardial atrioventricular cushions, producing a common atrioventricular orifice and deficiencies of the adjacent septal structures (Fig. 52.22B). The common orifice is guarded by a basically common valve, with superior and inferior leaflets bridging the scooped-out ventricular septum and tethered in both right and left ventricles. Although the left component of the valve thus formed is often interpreted as a 'cleft mitral valve', it bears no resemblance to the normally structured mitral valve, in reality, because it has three leaflets; the 'cleft' forms the zone of apposition between the left ventricular components of the bridging

leaflets. Two major subgroups are identified. The more frequent pattern has a common atrioventricular orifice, and the potential for shunting through the septal defect at both atrial and ventricular levels (see Fig. 52.22B). The less common form has separate right and left atrioventricular orifices, and shunting occurs only at atrial level. Occasionally, when the bridging leaflets are attached to the underside of the atrial septum, shunting is confined at ventricular level and is typically small.

Ventricular septal defects

The most common defect of the ventricular septum occurs around the expected site of the membranous septum in the right wall of the aortic vestibule, below the zone of apposition between the non-coronary and right coronary leaflets of the aortic valve (Fig. 52.22C). The defect is closely related to the septal leaflet of the tricuspid valve, but can extend to open into the ventricular outlet beneath the supraventricular crest. It occurs because the ventricular septum is incompletely closed by its membranous component, and it is often associated with overriding of the crest of the muscular septum by the aortic orifice, together with pulmonary stenosis or atresia and hypertrophy of the right ventricle (this latter combination is known as Fallot's tetralogy). Rarely, the pulmonary trunk can be normal or even dilated with this combination of abnormalities (Eisenmenger complex). Perimembranous defects, which retain the remnant of the membranous septum as part of their perimeter, can also occur with abnormal ventriculo-arterial connections, e.g. the pulmonary trunk may override the muscular septum (the Taussig-Bing anomaly). In perimembranous ventricular septal defects, the atrioventricular bundle and its right and left branches are always found along the posteroinferior margin of the defect.

Less commonly, a septal defect can be found in the ventricular outflow tracts roofed by the conjoined facing leaflets of the aortic and pulmonary valves. Such juxta-arterial defects are doubly committed, in that they open beneath the orifices of both aortic and pulmonary valves. They are the result of the failure of formation of the free-standing subpulmonary muscular infundibulum, but with appropriate septation of the ventriculo-arterial junctions. They usually have a muscular posteroinferior rim, which protects the atrioventricular bundle, but can extend to become perimembranous.

The third type of ventricular septal defect occurs within the musculature of the septum; defects may affect all parts of the septum and can be multiple, producing a so-called 'Swiss cheese' septum.

Defects within the inlet part of the septum are important because the atrioventricular bundle passes in their upper border. Perimembranous defects open into the inlet of the right ventricle, where the atrioventricular bundle is posteroinferiorly located.



Fig. 52.23 This heart possesses a common arterial trunk, with a common truncal valve overriding a juxta-arterial deficiency of the ventricular septum, the result of failure of septation of the arterial pole of the developing heart. (Courtesy of Professor RH Anderson, Institute of Child Health, University College, London. Specimen prepared by Dr Leon M Gerlis.)

ANOMALIES OF THE GREAT ARTERIES AND VEINS

A number of outflow anomalies may occur if the outflow cushions either fail to develop or fuse in inappropriate fashion. When the spiral septum cushions fail to fuse, the result is a common arterial trunk, represented by an undivided arterial channel, guarded by a common arterial valve, positioned above and astride the free margin of the muscular ventricular septum (Fig. 52.23). There is, therefore, a coexisting juxta-arterial deficiency of the ventricular septum. The right and left pulmonary arteries usually arise via a confluent segment but can take independent origin from the common arterial trunk, which continues as the ascending aorta. The common valve usually has three leaflets but may have two, four or more. The lesion is almost certainly linked to abnormal migration of cells into the heart from the neural crest.

Transposition of the arterial trunks (also referred to as transposition of the great vessels) is the condition in which the aorta arises from the right ventricle and the pulmonary trunk from the left. Better described as having discordant ventriculo-arterial connections, such hearts can coexist with deficiencies of cardiac septation. They can also be found with discordant connections at the atrioventricular junctions, producing congenitally corrected transposition. The developmental history of discordant connections remains unknown.

Double outlet ventricle exists when the greater parts of both arterial valves are attached within the same ventricle, almost always the right. For the circulation to continue, it is then necessary for the ventricular septum to be deficient, although the septal defect can rarely close as a secondary event. The position of the septal defect serves for subclassification. It is usually beneath the aorta or the pulmonary trunk, but can be doubly committed or even non-committed.

Either the systemic or pulmonary veins can be anomalously connected. Right and left superior vena cavae are present in some animals and occasionally persist in mankind. The most common systemic anomaly is a persistent left superior vena cava draining into the right atrium through the enlarged orifice of the coronary sinus. More rarely, the persisting left-sided vena cava may connect directly with the superior aspect of the left atrium, usually associated with so-called 'unroofing' of the coronary sinus, so that the orifice of the sinus functions as an interatrial communication. The most common lesion of the inferior vena cava is when its abdominal course is interrupted, with drainage to the heart via the azygos or hemiazygos venous system. This lesion is found most frequently with isomerism of the left atrial appendages. The pulmonary veins can be connected to an anomalous site individually or in combination. Totally anomalous connection is of most significance. Usually, the veins form a confluence behind the left atrium, which then connects to the superior vena cava, the coronary sinus or the portal venous system after traversing the diaphragm.

A right aortic arch is found most frequently with Fallot's tetralogy or with a common arterial trunk. It can also exist, together with a left arch, in various combinations known as arterial rings, which compress the

oesophagus, giving so-called dysphagia lusoria. Persistent patency of the ductus arteriosus must be distinguished from delayed closure. The persistently patent ductus can be an obligatory part of the circulation when associated with aortic or pulmonary atresia. Coarctation of the aorta can be found as an isolated lesion when the ductus arteriosus is closed, or with an open duct, when it is more likely to be associated with additional lesions within the heart.

In the adult, the right subclavian artery occasionally arises from the arch of the aorta distal to the origin of the left subclavian artery, and then passes upwards and to the right, behind the trachea and oesophagus. This condition is possibly explained by the persistence of the embryonic right dorsal aorta and the obliteration of the fourth aortic arch of the right side.

Ultrasound antenatal imaging of the heart and vessels

In a four-chamber view of the fetal heart, up to about 28 weeks' gestation, the heart occupies about one-third of the thorax in the anterior left quadrant of the chest. The two atria are of equal size, as are the two ventricles, which have an intact interventricular septum between them; the apex of the heart points to the left anterior thorax. In moving images, the ventricles show equal contraction, the two atrioventricular canals open equally, and the foramen ovale flap valve can be seen flickering in the left atrium (see Video 14.1). The blood flow through the heart can be demonstrated with colour flow (Fig. 52.24). After 28 weeks' gestation, the right ventricle may look dilated compared with the left.

Ultrasound evaluation of the fetal heart checks that the fetal heart and stomach are both on the left and that the liver is on the right. Within the four-chamber view, the function and rhythm of the heart and the origin of the great vessels from their respective ventricles can be observed. The great vessels can be assessed in transverse section and the aorta can also be assessed longitudinally (see Fig. 14.4); it normally arises in the centre of the thorax, and the vessels to the head and neck can be seen arising from the curve of the aortic arch. For further details of ultrasound findings, consult [Coady and Bower \(2015\)](#).

NEONATAL ARTERIAL AND VENOUS VESSELS

In the neonate, the blood vessels of the trunk and their associated visceral branches are relatively larger than the vessels in the limbs, an arrangement that favours central pooling of blood. Neonatal peripheral vessels are nearly microscopic; consequently, their cannulation poses much more of a problem than is the case with their adult counterparts. Large vessels are in the same relative positions as in the adult but may correspond to different vertebral levels. Thus, although the bifurcation of the common carotid artery into internal and external carotid arteries occurs at the level of the upper border of the thyroid cartilage, in both neonate and adult, the thyroid cartilage is relatively higher in the neonatal neck than it is in the adult. The renal arteries also arise higher in the neonate, often between T12 and L1, whereas they arise at the upper border of L2 in the adult. The abdominal aorta bifurcates into common iliac arteries at the upper border of L4, rather than at the lower border of L4, as occurs in the adult.

Central venous catheterization

Small-bore catheters can be fed into large central veins or into the right atrium via needles or catheters inserted in the peripheral veins. Typically, the median cubital or basilic veins are used in the upper limb, the long saphenous vein at the medial malleolus in the lower limb and the superficial temporal vein in the scalp. The tip of the catheter is sited at the entrance to the right atrium. The required catheter length is assessed from direct measurement of the distance between the point of surface entry in the limb to the right atrium, estimated at mid-sternal level. For details of umbilical catheterization, see [Chapter 60](#).

RESPIRATORY TREE

The development of the respiratory system can be conceptualized as the growth of six intertwined trees: the bronchial tree of conducting airways and terminal gas-exchanging sacs; the systemic arterial and venous trees that perfuse and drain the airways, respectively; the pulmonary arterial and venous trees that supply and drain the alveolar membrane, respectively, and anastomose with the systemic and venous trees (and are



Fig. 52.24 **A**, A normal four-chamber view at 22 weeks' gestation. **B**, The ventricular inflow with colour flow. Flow (seen in red) is seen from each atrium to each ventricle. Abbreviations: LA, left atrium; LV, left ventricle; MV, mitral valve; RA, right atrium; RV, right ventricle; TV, tricuspid valve. (Courtesy of Gurleen Sharland, Evelina London Children's Hospital.)

pivotal to the provision of blood oxygenation and carbon dioxide excretion after birth); and the lymphatic tree. Development of each tree is related to the other and all proceed in proximal to distal growth and expansion. Uniquely, the lung develops while not fulfilling its postnatal function, and must function efficiently immediately after birth or else the baby will require respiratory support and may die of respiratory failure. Current knowledge of the molecular basis of lung development is based on growth *in vitro* of human and animal explants (usually mouse and rat), and studies using knockout and transgenic mice (Herziges et al 2012). Anomalies of lung development make an important contribution to the development of airway diseases such as asthma and chronic obstructive pulmonary disease (COPD) (Bush 2008). The reader is referred to recent studies of lung developmental biology for more information (Roth-Kleiner and Post 2005, Kimura and Deutsch 2007, Maeda et al 2007, Bhaskaran et al 2009, Kho et al 2010, Morrissey and Hogan 2010, Sgantzis et al 2011, Ornitz and Yin 2012).

The development of the respiratory diverticulum can first be seen at stage 12 (approximately 26 days), when there is a sharp onset of epithelial proliferation within the foregut at regions of the endoderm tube destined to become the lungs, stomach, liver and dorsal pancreas. The specialist respiratory epithelium forms from the endoderm, whereas the other elements of the airway wall are of mesenchymal origin. The future respiratory epithelium bulges ventrally into the investing splanchnopleuric mesenchyme, then grows caudally as a bulb-shaped tube (Fig. 52.25; see Figs 52.1C, 60.3, 60.5). By stage 13, the caudal end of the tube has divided asymmetrically to form the future primary bronchi; with growth, the right primary bronchus becomes orientated more caudally, whereas the left extends more transversely. The trachea is clearly recognizable at stage 14. From this time, the origin of the trachea remains close to its site of evagination from the future oesophagus; however, longitudinal growth of the trachea causes the region of the future carina to descend, ultimately to lie within the thorax. Failure of complete separation between trachea and oesophagus will result in the baby being born with one of the variants of tracheo-oesophageal fistula (see below).

There is a sexual dimorphism in lung development. Androgens delay fetal lung maturation while stimulating fetal lung growth; in males, type II cells are less mature than in females, as they develop during the canalicular phase, perhaps because androgens block the effects of cortisol on fibroblast–pneumocyte factor concentrations. It is possible that, in male fetuses, surfactant production is delayed. The clinical counterpart is the observation that preterm males may have worse respiratory distress than females after control for other risk factors, such as degree of prematurity.

TRACHEA

The trachea starts to develop at stage 12, as a ventral outgrowth from the endodermal foregut into the mesenchyme that surrounds the sinus venosus and inflow tract of the heart (see Figs 52.1C, 60.3, 60.5). The point at which the original respiratory diverticulum buds from the foregut, the laryngotracheal groove, remains at a constant level during the embryonic period, and the trachea lengthens distally as the bifurcation point descends. The respiratory diverticulum generally becomes surrounded by angiogenic mesenchyme that connects to the developing sixth aortic arch artery and is essential for airway branching. By stage 17, the mesenchyme around the trachea is beginning to condense to form cartilage.

Initially, the tracheal mesenchyme is continuous with that surrounding the ventral wall of the oesophagus (see Fig. 52.3). Progressive lengthening and continued division of the tracheal bud, together with deviation of the lung buds dorsally, isolates the oesophagus and trachea within tissue-specific mesenchyme and facilitates regional differentiation, not only between trachea and lungs, but also within the lungs themselves, i.e. the number of lobes, or the degree of growth and maturity of a particular lung. Each lung develops by a process of dichotomous branching. For branching to occur, a cleft must develop in the tip (or side) of the epithelial tube. The epithelium then evaginates each side of the cleft, forming new branches that lengthen, and the process is then repeated (see Fig. 11.3). At the tips of the developing epithelial buds, the mesenchyme is flattened and densely packed, whereas it forms an ordered row of cuboidal cells along the side of the bud and in the clefts. Cells in both arrangements send processes towards the epithelial basal lamina, which is thicker in the clefts, but so attenuated as to be almost indistinguishable on the tips of the buds where the epithelium and mesenchymal cells form intimate contacts. Tenascin, an extracellular matrix molecule, is present in the budding and distal tip regions, but absent in the clefts. Conversely, fibronectin, an extracellular

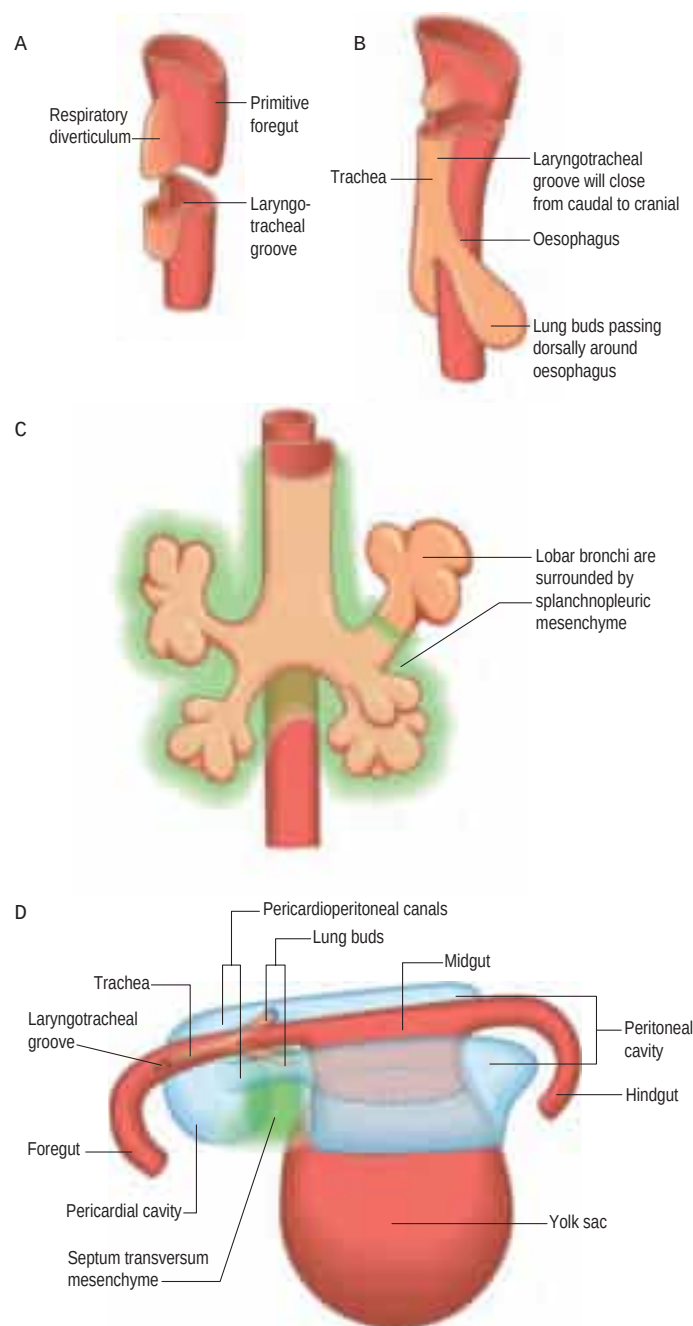


Fig. 52.25 Development of the respiratory tree and diaphragm.

A–C, Development of the endodermal respiratory tree. **D**, Major epithelial populations in the early embryo from a left dorsolateral view. The lung buds are bulging into the laterally placed pericardioperitoneal canals.

matrix molecule found commonly in basal laminae, is found in the clefts and along the sides of the developing bronchi, but not on the budding and distal tips. It is likely that, after the first generation, branching morphogenesis is not rigidly prespecified, but adapts to the space available within the mesenchyme (Blanc et al 2012).

The control of the branching pattern of the respiratory tree resides with the splanchnopleuric mesenchyme. Experimental recombination of tracheal mesenchyme with bronchial respiratory endoderm results in inhibition of bronchial branching, whereas recombination of bronchial mesenchyme with tracheal epithelium will induce bronchial outgrowths from the trachea. Experimental exposure of rat fetal airway to chick mesenchyme produces a chick airway branching pattern. Broadly, receptor tyrosine kinases and their ligands positively modulate growth and branching morphogenesis, while transforming growth factor beta (TGF- β) family members have an inhibitory effect.

Interestingly, even at this early stage, airway smooth muscle is innervated and contractile (Tollet et al 2002). Phasic contraction and relaxation of airways is important in growth factor release. Smooth muscle and nerves are found outside the airways at this developmental stage. Glial-derived neurotrophic factor (GDNF) is important in this process.

In the normal neonate, the trachea is relatively small in relation to the larynx (see Fig. 14.6B). The walls of the trachea are relatively thick

and the tracheal cartilages are relatively closer together than in the adult. The trachea begins at the upper border of the sixth cervical vertebra, a relationship that is conserved with growth, and it bifurcates at the level of the third or fourth thoracic vertebra.

Endotracheal intubation in the neonate

The insertion of an endotracheal tube is a procedure that may be required to resuscitate the newborn at birth and, subsequently, to enable artificial ventilation. Initially, the tube is usually introduced orally and it is then guided through the vocal cords under direct vision using a laryngoscope. The length of the trachea in the neonate can be as short as 3 cm in premature infants, and the distance from T1 to carina ranges from 1.4 cm in babies weighing 500–1000 g to 1.8 cm in those weighing 3001–3500 g. Once in place, the tip of the tube should be in the mid-trachea, well above the carina.

The required length of the tube can be estimated according to birth weight. If a shouldered tube is used, only the distal, tapered portion of the tube is inserted beyond the vocal cords, and the shoulders should prevent the tube being advanced too far. Confirmation of correct positioning of the endotracheal tube is obtained from a chest X-ray. The anatomical reference points used for the X-ray to assess the position of the endotracheal tube are the clavicles, the bodies of the vertebrae and the carina (although the last of these is not always visible on X-ray). Previously, it was advised that the tip of the endotracheal tube should be placed just below the clavicles, at the level of the first rib or 1–2 cm above the carina. Recently, this has been revised because positioning of the clavicles can vary according to angulation and placement of the baby, and the carina cannot always be identified. It is now suggested that the body of the first thoracic vertebra (T1) is a more stable reference point as the target for the tip of the endotracheal tube.

LUNGS

Branching of the conducting airways is generally complete by week 16. Airways subsequently increase in size but not in number. Alveolar development is largely (but not completely) a postnatal phenomenon. The development of pre-acinar blood vessels follows the conducting airway development, the airways acting as a template for blood vessel development by *de novo* vasculogenesis driven by epithelial-derived vascular endothelial growth factor (VEGF), whereas development of intra-acinar blood vessels follows the development of the alveolar ducts and alveolar sacs. Lung development is described histologically as progressing through embryonic, pseudoglandular, canalicular, saccular and alveolar phases. (For further details, see [Bush \(2005\)](#).)

Embryonic phase (0–7 weeks: appearance of lung buds and main pulmonary arteries)

The lung buds grow dorsally, passing each side of the relatively smaller oesophagus, and bulge into the medial walls of the laterally situated pericardioperitoneal canals (see Figs 52.25, 60.5A). The investing splanchnopleuric mesenchyme surrounding the lung buds contains a mixed population of cells. Further mesenchymal cells will differentiate into the smooth muscle cells that surround both the respiratory tubes and the blood vessels. In stage 13 embryos, proliferation of the adjacent splanchnopleuric coelomic epithelium (of the primary pleural cavities) is especially evident. The proliferative activity decreases in stage 14, and the mesenchyme becomes arranged in zones around the developing endoderm. Key molecules in airway branching include the fibroblast growth factors (FGFs), especially FGF-10, and their four receptors. There is almost certainly redundancy in the system, since mice null for either FGF-1 or FGF-7 exhibit normal lung development, despite the potent *in vitro* effects of these mediators. The key receptor appears to be FGFR2IIb, and those ligands that bind it. Sonic hedgehog is important in lung development; it is tightly regulated and over-expression leads to the formation of excessive lung mesenchyme. The three members of the gli family of transcription factors are implicated in Sonic hedgehog signalling. Sonic hedgehog produced in distal epithelial cells appears to act on mesenchymal cells to suppress FGF-10 to limit budding. Other signals interacting with this process include bone morphogenetic protein 4 and TGF- β 1.

At stage 15, angiogenic mesenchyme is apparent around the primary bronchi. It forms an extensive capillary network around each lung bud, receiving blood from the developing sixth aortic arch artery and draining it into an anastomosis connected to the dorsal surface of the left atrium in the mediastinal mesenchyme. The pulmonary arteries

arise from the sixth aortic arch arteries, and the pulmonary veins develop from a solitary channel derived from the dorsal mesocardium, which, by branching morphogenesis, establishes continuity with the vascular plexus formed in the mediastinal mesenchyme. The pulmonary veins become surrounded by myocardium to the level of the second bifurcation. The veins themselves expand and are incorporated into the roof of the left atrium; cardiac muscle is, therefore, found in the central branches of the pulmonary venous tree ([Hislop 2005](#)).

The lung buds on each side of the oesophagus project dorsally into the pericardioperitoneal canals at stage 15. After this stage, the coelomic epithelium at the perimeter of the lung surface follows a differentiation pathway to form the visceral pleura. Lobar or secondary bronchi can be seen at stage 16, and the bronchopulmonary segments are present at stage 17 (see Fig. 60.3). Later stages of respiratory development involve the repeated division of the bronchial tree to form the subsegmental bronchi.

Pseudoglandular phase (5–17 weeks: development of airways and blood vessels to level of acinus)

By stage 17 (late sixth to early seventh week), the separation of the lungs from the digestive system is complete and the pseudoglandular phase of pulmonary development, which includes the development of the lower conducting airways and the appearance of the acinar structures, can be identified. During this period, virtually the complete branching structure of the future bronchial tree is laid down, giving rise to 20 generations.

The growth and branching of the endoderm epithelium is controlled by the local investing splanchnopleuric mesenchyme. The airways begin to differentiate during this stage. Primitive ciliated cells appear at about week 7, initially in the region of the membranous trachea, and in the cartilaginous region by week 12. Ciliary biology and ciliopathy are an increasing focus of study ([Bush and Hogg 2012](#)). There are three known types of cilia. Best known are motile cilia, which propel mucus along epithelial surfaces, or are responsible for motility of unicellular organisms, but also probably have some cell signalling role; these are abnormal in primary ciliary dyskinesia (PCD, Kartagener's syndrome). Nodal cilia are also motile and are responsible for determining organ situs in the developing embryo ([Nonaka et al 2002](#)). Primary cilia are non-motile, are ubiquitous throughout the body, and have a wide range of signalling functions during development ([Goetz and Anderson 2010](#)). The exact *in utero* function of these early ciliated cells in the trachea is obscure. It is unclear whether airway ciliated cells develop from an undifferentiated precursor or through the further development of a population of secretory cells. The epithelium differentiates into ciliated and goblet cells. The proximal airways develop basal cells from week 11, and ciliogenesis is complete at birth in humans. Mucous glands develop by 12 weeks and enlarge in the submucosa; secretory activity has been identified in the trachea at 14 weeks. The splanchnopleuric mesenchyme condenses around the epithelium and differentiates into connective tissue cell types and smooth muscle, which differentiates proximal to the tips of the developing airways as they develop, from week 6 onwards. Smooth muscle cells are innervated from as early as week 8 (see below). First-trimester human tracheal smooth muscle cells have a fluctuating resting membrane potential that is associated with the spontaneous development of tone and peristalsis-like contractions of the airway of a myogenic nature, probably modified by neurohumoral factors. These contractions are important in moving fluid from distal to proximal within the airways, which is essential for normal lung development and may also stimulate mediator release. Clara cells develop in the peripheral airways during the pseudoglandular period. They produce a 10 kDa protein (CC10) with immunomodulatory and anti-inflammatory activities, as well as producing a surfactant apoprotein. Cartilage also develops during this period, and is found in the airways in an adult distribution by 24 weeks. By the end of this period, the airway branching and the pre-acinar vascular patterns are fully mature. A normal airway 'template' is essential for normal alveolar formation; in conditions characterized by an abnormal airway branching pattern (e.g. both ipsilateral and contralateral to a diaphragmatic hernia), a degree of pulmonary alveolar hypoplasia is inevitable.

Endothelial development is also seen in the pseudoglandular phase when capillary networks form around the developing lung buds. These networks will become capillary anastomoses around the future alveoli. The mesenchyme produces both the endothelium and the smooth muscle cells of the tunica media of the vessels. Vimentin occurs in the cells around developing vessels in the pseudoglandular stage, but is replaced by desmin in the saccular phase.

Canalicular phase (17–27 weeks: formation of respiratory airways and thinning of blood-gas barrier)

During the canalicular phase, about two to three generations of branching take place, after which the amount of mesenchyme around the branching tips of the dividing respiratory tree decreases and the distal airspaces widen. At 23 weeks, longitudinal sections of the future distal regions show a sawtooth margin, which may indicate the site of further acini. Peripheral growth is accompanied by an increase in the capillary network around the distal airspaces. In many places, the capillaries are in close contact with the respiratory cuboidal epithelium. The primitive cuboidal cells, which hitherto predominated in this part of the lung, differentiate into type 2 epithelial cells (pneumocytes), containing lamellar bodies that form the intracellular storage bodies of surfactant, and type 1 epithelial cells (pneumocytes). Type II pneumocytes are believed to be the stem cells of the alveolar epithelium. Apposition of the capillary networks to the thin pneumocytes (type 1), and reduction of the interstitial tissue of the lung, are prerequisites for future effective gas exchange. By 24 weeks of gestation, the histological characteristics of the airways, including cartilage distribution, are the same as in the adult.

Saccular/alveolar phase (28 weeks to term: first appearance of alveoli in humans)

Thin-walled terminal saccules are apparent at the saccular stage and will become alveolar ducts as development proceeds. The expansion of the prospective respiratory airspaces that occurs during this period is accompanied by a further decrease in the amount of interstitial tissue, and the capillary networks become ever more closely opposed to the pneumocyte epithelium. Invaginations (secondary crests) develop from the saccule walls. As a crest protrudes into a saccule, part of the capillary network becomes drawn into it. After the later expansion of the saccules on each side of the crest, a double capillary layer becomes annexed between what are now alveolar walls. During the saccular stage, elastin is deposited beneath the epithelium (an important step for future alveolar formation), and surfactant production from type 2 pneumocytes (essential for the survival of a preterm neonate) matures.

Maturation of the blood–gas barrier is an essential part of normal human development but the molecular basis of this is little understood. Mechanical stretch drives type 2 to type 1 cell differentiation. VEGF is a key mediator; it is upregulated by hypoxia, in late gestation, by cyclic stretch and by the transcription factor HIF2 α . Type 2 cell maturation and capillary development fail in HIF2 α ^{-/-} mice. VEGF also drives surfactant production from glycogen in type 2 cells.

Alveolar phase

Exactly when the saccular structure of the lung can be termed alveolar is not yet clear: estimates range from 28 to 32 weeks. The distal airspaces expand during late gestation and continue to do so after birth. The number of alveoli present at birth (variously estimated as between none and 5×10^7) is controversial but the numbers probably increase particularly rapidly over the first 6 months of life. This process is accompanied by fusion of adjacent capillary nets, so that, shortly after birth, there is an extensive double capillary net. Fusion of these layers is apparent at 28 days postnatally, extensive at 1.5 years and probably complete by 5 years. The alveolar stage is now considered to continue beyond infancy and to last throughout childhood (see below). Interestingly, in rats and mice, formation of alveoli is entirely postnatal.

The formation of millions of alveoli is accomplished by a complex process of folding and division. Existing walls of distal airspaces become thinner as pneumocyte epithelial cells flatten, and some cells undergo apoptosis. Ridges subsequently grow out from the sides of the saccule walls, forming primitive alveoli. The secondary septa contain a double capillary layer and further new alveoli form by the infolding of one of these layers, further subdividing the airspaces. The double capillary networks then undergo remodelling to form the familiar single capillary sheet around each alveolus. Conducting airways also undergo profound postnatal changes and exhibit increased smooth muscle and bronchoconstrictor responsiveness. The precise molecular signalling for these processes has yet to be established. However, they are clearly tightly regulated developmentally; hyperoxia or dexamethasone treatment of rats during the alveolarization period permanently abrogates alveolar development. This experimental finding implies that iatrogenic damage to the developing alveoli when ventilating a preterm baby with oxygen-

enriched mixtures is probably irreversible and, indeed, may be the basis of ‘new’ bronchopulmonary dysplasia.

Elastin gene expression, possibly modulated through retinoic acid, appears to be of pivotal importance in alveolar budding. Retinoic acid treatment of rats increases alveolarization, and salvages some alveoli in an experimental model, whereas mice with a deletion in the retinoic acid receptor γ have impaired alveolarization. Retinoic acid regulates the expression of a number of growth signalling molecules, homeobox genes and lung epithelial genes; other gene groups involved in alveolarization probably include TGF- β , platelet-derived and fibroblast growth factors. (For further reading on the control of alveogenesis, see Galambos and DeMello (2008) and Hind et al (2009).)

Development of the intrapulmonary vasculature and lymphatics

In an adult lung, the pulmonary arteries accompany the airways. The pulmonary veins have a similar number of branches but are separated from the airways by the alveoli; they have a different embryological origin, being derived from the mediastinal mesenchyme proximally rather than from the splanchnopleuric mesenchyme around the bifurcating respiratory tree. New vessels are formed by either vasculogenesis (the formation *in situ* of new blood vessels) or angiogenesis (outsprouting from existing vessels). Failure of separation of pulmonary veins from the arterial–airway bundle is a feature of the alveolar–capillary dysplasia spectrum (alveolar–capillary dysplasia with misalignment of the pulmonary veins), and may be caused by *FOXF1* or *STRA6* mutations (Bishop et al 2011). The mesenchyme around the single lung bud contains a number of cells that stain positively for the endothelial cell markers CD34 and CD31. The earliest pulmonary vessels form in the mesenchyme by vasculogenesis; the capillaries coalesce to form small blood vessels alongside the airways. By 34 days of gestation, blood circulates from the aortic sac via pulmonary arteries into a capillary plexus around the two lung buds and drains to the developing atrium. As each new airway forms in the mesenchyme, a new plexus forms as a halo around it and coalesces with the vessels already alongside the previous airway. In this way, addition of the newly formed tubules to the existing vessels is sustained; the airways act as a template for the development of blood vessels. Vessel formation occurs at least until the end of the pseudoglandular stage and is probably controlled by production of VEGF from the epithelial cells of the peripheral lung buds. As the arteries increase in size, they acquire a muscular wall. Initially, this arterial tunica media is derived from the bronchial smooth muscle of the adjacent airways, but as the vessels progressively enlarge, cells derived from the splanchnopleuric mesenchyme differentiate around the wall to form layers of smooth muscle cells. The muscular wall is thick relative to the lumen and this, in part, increases resistance to blood flow, although only 10% of blood flow goes through the pulmonary circulation of the lung during intrauterine life. None the less, this small blood flow is important in normal lung development (see below). At 20 weeks, the structure of the pulmonary vessels is the same as it is in the adult. Vascular innervation follows muscularization; the vasoactive peptides in the nerves are predominantly vasoconstrictor.

As the pulmonary veins enlarge, they become separated from the airways by lymphatic vessels lying within connective tissue. At around week 12, the peripheral veins develop a single layer of smooth muscle cells in their walls. This layer is derived only from the splanchnopleuric mesenchyme and not from bronchial smooth muscle (as occurs in the arteries). The lumen of each vein is relatively large and the wall is relatively thin at all levels. Arteries and veins continue to develop in the canalicular phase, probably by angiogenesis; dividing cells are seen in the peripheral capillaries. The epithelium of the most peripheral conducting airways flattens when the developing capillaries come to lie immediately subjacent to it during this phase. (For a review of the development of the pulmonary circulation, see Hislop (2005).)

Lymphatic capillaries and vessels run within bronchovascular bundles. Lymphatic channels develop from outgrowths of vascular endothelial cells; the molecular trigger for lymphatic sprouting is VEGF-C and D, acting through the VEGF receptor 3 (Janér et al 2006). (See Kulkarni et al (2011) for a review of lymphatic development in the lung.)

As well as developing a pulmonary circulation, the lung also develops a bronchial supply from the systemic circulation. Two or three bronchial arteries grow from the descending aorta from week 8 onwards, entering the lung at the hilum. These arteries extend down the intrapulmonary airways, eventually forming subepithelial and adventitial plexuses, which, by birth, have reached the distal portions of the bronchioli.

The bronchial veins from the periphery of the lungs drain into the pulmonary veins. Bronchial arteries also supply the pleura near the hilum, and form vasa vasorum in the adventitia of the large arteries and veins. Postnatally, inflammatory lung conditions, such as asthma, cystic fibrosis and bronchiectasis, cause hypertrophy of the bronchial circulation and these vessels may bleed, sometimes giving rise to massive haemoptysis (see [McCullagh et al \(2010\)](#)).

Intrauterine maturation of the lungs

There are more than 30 different cell types within the adult lung that mature during development. Only a few important changes relevant to human disease will be highlighted here.

Pulmonary surfactant is produced by type 2 alveolar epithelial cells and stored in lamellar bodies. Surfactant proteins (Sp) B and C are surface-active, whereas A and D are part of the collectin family of pattern recognition receptors. Surfactant is crucial for maintaining the functional integrity of alveoli; the main morbidity of extreme prematurity reflects surfactant deficiency causing neonatal respiratory distress (treatment of these babies has been transformed by the availability of exogenous surfactant). Mutations in the genes encoding *SpB* and *SpC*, *ABCA3* (responsible for surfactant processing) and the transcription factor *TTF-1* (controlling *SpB*, *SpC* and *ABCA3* expression) can all present in the newborn period in term babies with relentlessly progressive respiratory distress, with diffuse ground-glass shadowing on imaging studies ([Bush and Nicholson 2008](#), [Galambos et al 2010](#)). Fetal breathing movements appear to be important in surfactant synthesis: the functional maturation of the surfactant system can be accelerated therapeutically by the administration of steroids to the mother.

Normal lung development requires sufficient intrathoracic space, normal fetal breathing movements and sufficient amniotic fluid. There is evidence that perturbation of fetal lung blood flow also affects lung growth, even though most of the right ventricular output of the heart is shunted away from the lungs through the ductus arteriosus; pulmonary valve stenosis is associated with pulmonary hypoplasia.

Although many fetal organs are able to grow to normal proportions even if they are in abnormal locations, this is not the case for the lungs. Lung growth becomes impaired by restricted expansion, and distension of the developing lung may provide a major stimulus to growth during normal development. Absence or impairment of fetal breathing movements, and anomalies affecting diaphragmatic activity, are all associated with pulmonary hypoplasia. It is believed that normal fetal breathing movements increase the lung volume and stimulate growth of the distal airspaces. These movements involve rhythmic activation of the diaphragm and the muscles of the upper respiratory tract; even though they are necessarily very small compared with those seen after birth (because the fetal airways are filled with lung fluid), there is evidence that these phasic movements are important in the release of growth factors. (The role of fetal breathing movements has been reviewed in [Inanlou et al \(2005\)](#).)

During development, the mucous glands of the trachea and bronchi secrete a chloride-rich fluid, which usually passes up the respiratory tract to mix with the amniotic fluid. The relationship between lung fluid and amniotic fluid is far more complex than was previously believed. Pulmonary hypoplasia at birth may be associated with severe congenital urinary obstruction and oligohydramnios (Potter's syndrome). In renal agenesis, reduced bronchial branching occurs as early as 12–14 weeks of gestation (i.e. at a time before amniotic fluid is produced by the kidneys), which suggests that a direct renal factor supports lung development. Later, the presence of amniotic fluid is necessary for normal fetal lung development. The fetal lung is a net fluid secretor, the output of fluid reaching as high as 5 ml/kg shortly before birth (see [p. 179](#)). Most of the fluid produced within the lungs remains there because of the mechanical effect exerted by amniotic fluid pressure and, normally, only a small amount of this fluid contributes to the amniotic fluid. The normal functioning of the kidneys regulates the volume and pressure of the lung airway fluid and may, in turn, provide the pressure needed for expansion and enlargement of the bronchial and pulmonary systems. Interestingly, obstruction to the fetal airway causes accelerated maturation of alveoli; this approach has been used therapeutically in congenital diaphragmatic hernia, where intermittent inflation and deflation of an intratracheal balloon in a lamb model resulted in better lung growth and maturation ([Nelson et al 2005](#)).

Two other cell types that are part of normal lung development, neuroendocrine cells and glycogen-containing cells, are implicated in postnatal paediatric interstitial lung disease, with early onset of respiratory distress. Neuroendocrine cells are found in the normal developing airway. Neuroendocrine cell hyperplasia of infancy (NEHI) is character-

ized by persistence of bombesin-positive neuroendocrine cells. Persistence of glycogen-containing cells results in pulmonary interstitial glycogenosis (PIG), which is not related to any of the systemic glycogen storage diseases. Both conditions tend to remit over time, and in neither case is it known whether the abnormal cells are of pathophysiological significance or are merely markers of another process.

Adverse effects on intrauterine lung development

There is increasing evidence from epidemiological studies that antenatal factors have effects on lung development that are long-lasting. The earliest major effect is congenital diaphragmatic hernia, which leads to both ipsilateral and contralateral anomalies in branching pattern in the first 16 weeks of gestation that cannot be corrected subsequently, even by intrauterine tracheal balloon occlusion. Inevitably, abnormal branching leads to abnormal alveolarization in this condition. The most important later influence is maternal smoking but there is evidence that maternal exposure to pollution is also important ([Latzin et al 2009](#)). Animal studies have confirmed that antenatal nicotine exposure leads to structural changes in the fetal lung ([Elliot et al 2001](#), [Sekhon et al 2002](#), [Fu et al 2011](#)). Other factors that may be important include maternal hypertension in pregnancy, and maternal antibiotic and paracetamol usage.

Ultrasound antenatal imaging of the lungs

Pulmonary hypoplasia, an absolute decrease in lung volume and weight for gestational age, may be identified but cannot be confirmed. It is normally associated with oligohydramnios and renal agenesis. Unilateral hypoplasia may be suggested by rotation and displacement of the heart on ultrasound examination. In bilateral pulmonary hypoplasia, the heart appears relatively large with reduced chest volume. However, intrauterine growth retardation may also cause similar findings. Increased echogenicity of the lungs indicates conditions that prevent the normal circulation of lung fluid (for further details, consult [Coady and Bower \(2015\)](#)). Details of prenatal lung anomalies and their detection are given in [Epelman et al \(2013\)](#).

FUNCTIONAL CHANGES IN THE LUNG AROUND THE TIME OF BIRTH

Just prior to birth, the placenta is the organ of respiration. Virtually all the venous return to the right heart is shunted away from the lungs through the foramen ovale and the ductus arteriosus. Pulmonary vascular resistance (PVR) is maintained at a very high level by the muscular precapillary pulmonary vessels. At birth, the umbilical cord is tied and cut; within seconds, the lungs must take over all respiratory function in order to maintain life. They also must convert from secreting fluid to absorbing it. The mechanisms that mediate these dramatic changes are obscure. However, it is clear that the first breath results in vasodilation by at least two mechanisms. First, there is the mechanical effect of traction on the vasculature: as the chest wall expands, it pulls open the pulmonary vessels. Second, oxygen entering the lungs for the first time produces pulmonary vasodilation. The experimental demonstration that PVR falls, even if the fetus is delivered into an atmosphere of pure nitrogen, implies that other mediators, including cyclo-oxygenase metabolites, are important. The gene for cyclo-oxygenase 1 (but not 2) in endothelium and vascular smooth muscle shows enhanced expression in late fetal and early postnatal life. Endothelin receptor expression increases around the time of birth, implying a role for this system in postnatal adaptation ([Levy et al 2005](#)). The role of nitric oxide in postnatal adaptation is controversial and there are likely to be important species differences. Nitric oxide synthase (NOS) is more abundant in young compared with mature animals, and smooth muscle sensitivity to nitric oxide may be greater at birth than in older animals. The effects of mediators may differ ante- and postnatally, e.g. endothelin (ET)-1 causes vasodilation in fetal sheep, but vasoconstriction postnatally in lambs. Immediately after birth, and before structural remodelling has taken place, PVR may rise steeply if the baby becomes even minimally hypoxic or acidaemic. The fetal shunts (foramen ovale and ductus arteriosus) reopen and right to left shunting causes profound hypoxaemia and 'persistent fetal circulation'.

To summarize, at birth, the lung volume is small but related to body weight; all airways are present and differentiated (cartilage, glands, muscle, nerves); the one-third to one-half of the adult complement of alveoli present at birth give sufficient surface area for gas exchange; the

morphology of the blood–gas barrier is the same as in adults; and most arteries and veins are present.

POSTNATAL LUNG DEVELOPMENT

The amount and type of connective tissue in the lung change after birth. The neonatal lung has abundant type III and type IV collagen, but little type I collagen. Types III and IV collagen are not mechanically strong, suggesting that the neonatal lung has a phenotype that facilitates the changes in cell shape and orientation that characterize adaptation to extrauterine life. It is probable that the rapid deposition of type I collagen postnatally contributes structural stiffness to the blood vessel walls.

Lung volume increases most rapidly during the first year of life and also increases more rapidly than airway calibre during this time, a finding consistent with the concept of dysanaptic lung growth, i.e. a dissociation between the growth of the lung parenchyma and the airways. Until recently, it was thought that most of the alveoli were formed by 2 years of age, and that subsequent lung growth primarily resulted from an increase in the size of individual alveoli. However, recent work measuring alveolar size throughout childhood and adolescence using hyperpolarized helium has shown that neo-alveolarization, as well as increase in alveolar size, continues through childhood and adolescence (Narayanan et al 2012). This has important implications for diseases characterized by neonatal arrest of alveolarization, e.g. after ventilation of the extreme preterm baby. Forced vital capacity is reported to be the same in male and female infants and very young children (Jones et al 2000), which is consistent with postmortem morphometric measurements. By the second year of life, the rate of increase in forced vital capacity is similar to that reported in older children.

During childhood, lung volumes and flow rates increase linearly with height, with a greater intercept and more positive slope in boys compared to girls (Stanojevic et al 2008). The linearity of these changes is interrupted by puberty (Rosenthal et al 1993a, 1993b, Rosenthal and Bush 2002), when important shape changes occur in the thorax, particularly in boys; these are mirrored by changes in the size of the underlying lung (and, in particular, the alveoli), as manifested by an abrupt increase in vital capacity, forced expired volume in 1 second, and total lung capacity. Thereafter, the change in lung size again proceeds in a linear fashion. High-resolution computed tomographic scanning has been used to measure airway dimensions (de Jong et al 2006), although this technique should be used with caution because of the radiation exposure.

There is little in the way of structural data to set alongside epidemiology. The reticular basement membrane (lamina reticularis) is first detected at about 30 weeks' gestation, thickens throughout childhood and adolescence, and plateaus, or possibly declines, in thickness with age (Tsartsali et al 2011). In terms of airway smooth muscle, autopsy data suggest that it increases over the first 2 years of life; limited data from endobronchial biopsies suggested that it did not change from age 3 years throughout childhood (O'Reilly et al 2013). Goblet cells also increase over the same timeframe (Hislop and Haworth 1989).

Normal postnatal pulmonary arterial development

Immediately after birth, dramatic remodelling of the pulmonary vasculature occurs, to effect an abrupt reduction of pulmonary vascular resistance. This process continues at a rapid rate throughout the first 1–2 months, while the lungs adapt to extrauterine life, and then more slowly throughout childhood. Failure to remodel in the presence of an anatomically normal heart leads to persistent pulmonary hypertension. Normal postnatal pulmonary arterial development in the full-term neonate can be divided into three stages.

Stage one This lasts from birth to about postnatal day 4 and concerns the immediate adaptation to extrauterine life. At birth, the endothelial cells of the precapillary arteries are squat and have narrow bases on the subendothelium, a low surface to volume ratio and many surface projections. Five minutes after birth, the endothelial cells are thinner and gradually show less cell overlap, the surface to volume ratio increases, few cell projections are seen, the vessel wall becomes thinner and the luminal diameter increases (Fig. 52.26). The smooth muscle cells show a significant reduction in diameter during this time.

Stage two This lasts from around day 4 to 3–4 weeks and is the time when the cells deposit matrix around themselves to fix their new

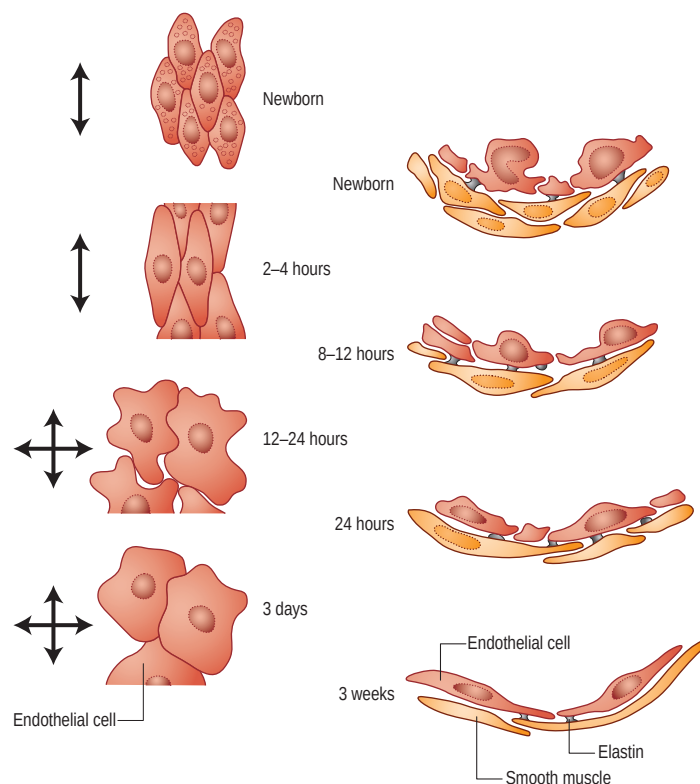


Fig. 52.26 *En face* views (left) and transverse sections (right) showing the changes in the endothelial and smooth muscle cells of small muscular pulmonary arteries accompanying terminal bronchi from the neonatal period to 3 weeks after birth. (With permission from Haworth SG 1992 Pathophysiological and metabolic manifestations of pulmonary vascular disease in children. Herz 17(40:254–261).)

positions. At birth, the internal elastic lamina of the small muscular arteries consists only of amorphous elastin in a basal lamina-like matrix. By 3 weeks of age, a definitive elastic lamina is evident, although it is heavily fenestrated, permitting contact between the endothelial cells and the smooth muscle cells.

Stage three This continues into adulthood. The intrapulmonary arteries increase in size and their walls increase in thickness. However, the maturation of all of the pulmonary vascular smooth muscle cells, from the hilum to the precapillary bed, is not advanced until 2 years. As the distal airspaces expand, the capillary nets fuse from one alveolus to another, forming, for a period, an extensive double capillary net; this process can be seen from postnatal day 28, becomes more extensive by 1.5 years, and is believed to be complete by 5 years.

Adverse effects on postnatal lung development

The long-term effects of preterm birth and its treatment are changing over the years. Current extremely premature babies, who are treated with surfactant and ventilated using low pressure, high-risk strategies, have arrest of alveolar development and pulmonary hypoplasia. Survivors from the pre-surfactant era have disease dominated by airflow obstruction. Importantly, even so-called 'late pre-term' babies (born at 33–34 weeks' gestation) have persistent airflow obstruction (Kotecha et al 2012). Postnatal lung development is also impaired by exposure to tobacco smoke, and there is increasing evidence that air pollution (Schultz et al 2012) and early sensitization to aeroallergens (Illi et al 2006) also have an adverse impact on normal lung growth. The first 4–6 years of life seem to be a key time window, after which there is no catch-up growth in lung function (Oswald et al 1997, Morgan et al 2005, Stern et al 2007); early effects on lung growth have lifelong implications.

CONGENITAL MALFORMATIONS OF THE TRACHEA, BRONCHI AND LUNGS

This is an enormous topic and only a small number of anomalies can be mentioned here (for further information, see Abel et al (2012)). The concept of the lung as six 'trees' has been mentioned earlier; all but the

systemic venous tree may contribute to a congenital thoracic malformation (CTM) (Bush 2009). In addition, congenital anomalies of the heart and great vessels, and of the chest and abdominal walls, including neuromuscular disease, may impact on lung development. It is somewhat artificial to describe airway malformations in isolation, and the possibility of associated vascular anomalies must always be considered. Moreover, descriptions of what is seen clinically should be kept separate from speculations about the embryological origins of the anomaly.

Disorders of the proximal airways



Available with the *Gray's Anatomy* e-book

Disorders of the distal airways



Available with the *Gray's Anatomy* e-book

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Bonus e-book images and video

Fig. 52.4 A Morgagni hernia in an 8-year-old boy, as displayed at laparotomy.

Fig. 52.5 A plain radiograph of the chest and abdomen in a 1-day-old neonate with left posterolateral congenital diaphragmatic hernia.

Fig. 52.24 A, A normal four-chamber view at 22 weeks' gestation. **B**, The ventricular inflow with colour flow.

Fig. 52.27 Examples of tracheo-oesophageal fistulae and oesophageal atresia.

Video 52.1 Animation of the pattern of contraction of the early heart tube.

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Tracheomalacia and bronchomalacia

Anomalies in cartilage development lead to the affected airways being 'floppy' and collapsing during inspiration or expiration, depending on whether they are outside or inside the bony thorax, respectively; these anomalies are termed tracheomalacia when the trachea is affected, and bronchomalacia when the bronchi are involved. The anomaly may involve a small, localized area or may be more generalized (e.g. Williams–Campbell syndrome, a congenital deficiency of the airway cartilage, in which there is diffuse bronchomalacia from the second to the seventh generation of bronchi). Tracheobronchomalacia usually presents in early infancy with cough, tachypnoea, stridor and wheeze; it may also be associated with cardiac or respiratory anomalies, such as absent pulmonary valve syndrome, or tracheo-oesophageal fistula, respectively. Patients with significant obstruction and apnoea may be treated by the insertion of airway stents if the disease is localized, although there is only limited long-term experience in children with stents; generalized disease may mandate respiratory support, either with non-invasive mask ventilation during sleep, or via a tracheostomy, depending on severity. Surgical treatment by aortopexy is controversial.

Tracheo-oesophageal fistulae

Tracheo-oesophageal fistulae are the most common anomalies of the lower respiratory tract and occur in about 1 in 3000 births. They usually present in the newborn with recurrent respiratory distress and choking spells, although late presentation in adult life has been described and may be associated with oesophageal atresia. Normally, the oesophagus lengthens with the rapid elongation of the embryo up to 7 weeks. The rapid proliferation of the oesophageal lumen partially obliterates the mucosa and recanalization is not complete until 8 weeks. The cellular processes that lead to separation of the trachea and oesophagus occasionally produce oesophageal atresia. Tracheo-oesophageal atresia is rare, with an incidence of 1 in 3000 to 1 in 4500 births. Five types of tracheo-oesophageal fistulae may be recognized (Fig. 52.27). In almost all cases, the oesophagus ends blindly and the stomach is connected to the lower end of the trachea. Because of this connection, the abdomen becomes rapidly distended with air once the baby is delivered and starts breathing. Prenatally, polyhydramnios may be a clinical feature but may not be apparent until the third trimester. In two of the five types of tracheo-oesophageal fistulae, there is no communication between the stomach and the upper gut. Such cases are identifiable on ultrasound because the stomach should always be visible at a 20-week examination; its absence on ultrasound should prompt further evaluation.

Tracheo-oesophageal fistula is commonly associated with other congenital anomalies, including cardiovascular defects (30%), anorectal anomalies (15%) and genitourinary anomalies (15%). These defects may be combined in the VATER, VACTER and VACTERL associations, which involve varying combinations of vertebral defects, anorectal atresia, cardiac anomalies, tracheo-oesophageal fistula and oesophageal atresia, together with radial aplasia and other limb anomalies, renal anomalies and the presence of a single umbilical artery.

Complete cartilage rings

The normal large-airway cartilages are horseshoe-shaped, and the ends of the horseshoes are bridged by the membranous part of the trachea. Complete cartilage rings may develop in the trachea and large airways, and the posterior, membranous, part of the airway may be absent in a single cartilage or in long segments of the airway. This anomaly may be associated with vascular anomalies such as pulmonary artery sling (where the left pulmonary artery originates from the right pulmonary artery and not from the main pulmonary trunk). Extensive severe disease may present with neonatal respiratory distress; later presentations include apparently steroid-resistant asthma. Tracheal transplantation has been used to treat long-segment tracheal stenosis. Tracheal cartilage shape is abnormal in cystic fibrosis in both human infants and the experimental pig model, but the cartilages are not continuous rings in this condition (Meyerholz et al 2010).

Agenesis and aplasia of the lungs

Agenesis and aplasia of the lungs are extremely rare congenital anomalies, in which there may be unilateral or bilateral absence of the lung. In agenesis, the airway stump is absent, i.e. the trachea is absent in bilateral disease, or the primary bronchus is absent in unilateral disease. In aplasia, there is a rudimentary airway stump but no distal lung; the contralateral lung is enlarged, with a greater number of alveoli, but has a normal bronchial branching pattern. Individuals with aplasia may present with recurrent infection, dyspnoea and reduced exercise capacity, and there may be pooling of bronchial secretions, with secondary infection and overspill of infected secretions into the contralateral normal lung.

Bronchogenic and other cysts

Cysts of various types, by order of frequency, may be found in the carinal (51%), right paratracheal (19%), para-oesophageal (14%), hilar (9%) and pericardial, retrosternal and paravertebral (7%) regions. Clinically, they are part of the spectrum of congenital thoracic anomalies, and can only be classified in more detail pathologically when the wall of the cyst is examined; they may have a systemic blood supply and venous drainage. Multicystic anomalies, described as various types of congenital cystic adenomatous malformations (CCAM) and sequestrations (intra- and extrapulmonary), are best considered generically as a CTM spectrum, at least until examined pathologically; even then, features of CCAM and sequestration may be found in the same anomaly, rendering classification difficult. Presentation may involve a range of problems, from intractable newborn respiratory distress to a chance finding in adult life. If the infant is symptomatic, then surgical resection is indicated. Treatment of the asymptomatic anomaly is controversial; some centres would operate electively on all but the tiniest malformations, whereas others would manage even large, asymptomatic, cystic malformations conservatively. Evidence is summarized in a recent meta-analysis (Stanton et al 2009).

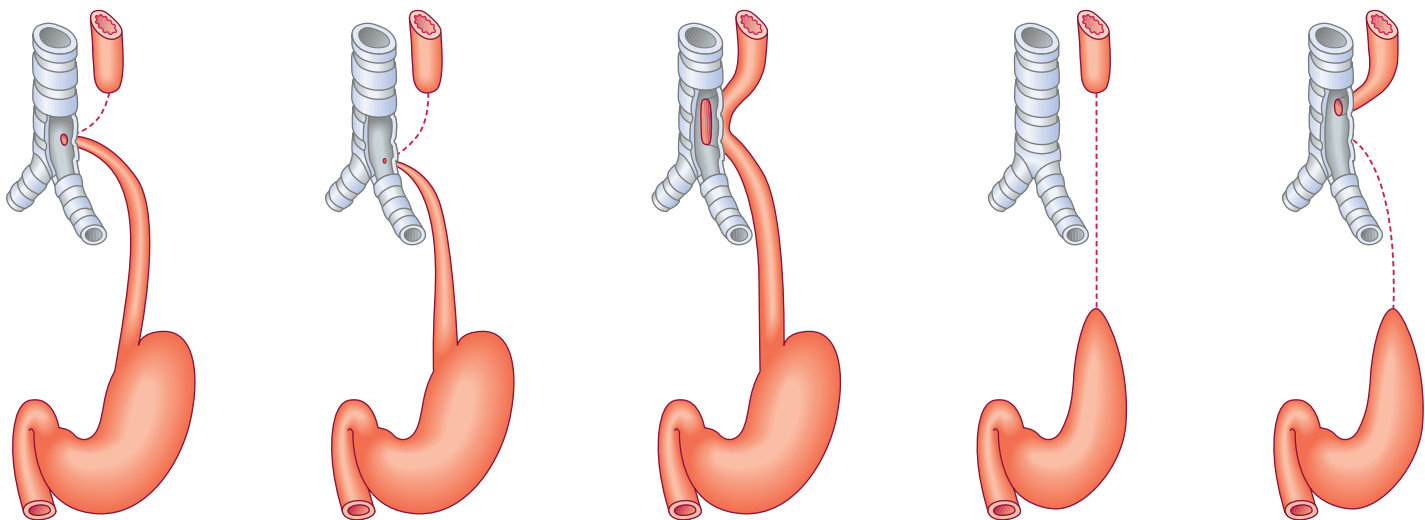


Fig. 52.27 Examples of tracheo-oesophageal fistulae and oesophageal atresia.

Bronchial atresia

Bronchial atresia is a congenital condition in which a major bronchus or segmental airway ends blindly or with a thin membrane. Pulmonary development normally occurs distally and, over time, this tissue becomes distended with debris and mucus (bronchocele), with adjacent over-inflated lung tissue. Typically, bronchial atresia affects the left upper lobe (64% of cases); the chest radiographic findings are of a perihilar ovoid density with strands projecting into a localized area of hyperlucent lung. The left lower lobe is affected in 14% of cases and the right lower and right middle lobes in 8% of cases. The majority of

cases are asymptomatic and are revealed as incidental findings on chest radiographs. However, many eventually become the seat of recurrent infection and require operative removal. Some infants may show a congenital large hyperlucent pulmonary lobe (CLHL). This condition is termed 'congenital lobar emphysema', although it has nothing to do with adult emphysema. Presentation in severe cases is with newborn respiratory distress; in this case, early surgery is required. Otherwise, it may be a chance radiographic finding later in life; in this case, no action is needed.

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